

ONE OF $\zeta(r+2), \dots, \zeta(6r-1)$ IS IRRATIONAL FOR EVERY ODD r , AND COMPLETE PROOFS FOR $\zeta(7), \dots, \zeta(21)$ AND $\zeta(9), \dots, \zeta(33)$, VIA ZUDILIN'S HIGHER-DERIVATIVE CONSTRUCTION

ABSTRACT. We prove: (A) at least one of the eight numbers $\zeta(7), \zeta(9), \dots, \zeta(21)$ is irrational; (B) at least one of the thirteen numbers $\zeta(9), \zeta(11), \dots, \zeta(33)$ is irrational. These sharpen Zudilin's theorems of 2001–2002 (one of $\zeta(7), \dots, \zeta(35)$ and one of $\zeta(9), \dots, \zeta(51)$ in [11]; 37 and 53 in [12]), which as far as we can find are the best published results of this shape. The linear forms are the ones of Zudilin's general construction (*J. Théor. Nombres Bordeaux* 16 (2004), §8) with $r-1=4$, resp. 6, derivatives; the arithmetic of the forms is his Lemma 19, which is valid for every odd r . What was missing for $r \geq 5$ is the analytic lemma (his Lemma 20, proved there only for $r=3$). We supply it in a form adapted to computer verification: the size of the form is governed by the Fourier mode $e^{\pm i(r-2)\pi t}$ of $\sin^r(\pi t) \cot_r(\pi t)$, whose saddle point is certified together with an explicit steepest-descent path, and a phase condition gives non-vanishing. All finitely many numerical inequalities are listed with the error control used to verify them. The same proof re-derives Zudilin's “one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ ” from a slightly different direction vector, as a consistency check. Running the same certificates for more derivatives gives (Section 7) at least one irrational number in each of $\{\zeta(11), \dots, \zeta(45)\}$, $\{\zeta(13), \dots, \zeta(55)\}$, $\{\zeta(15), \dots, \zeta(65)\}$, $\{\zeta(17), \dots, \zeta(75)\}$ ($r=9, 11, 13, 15$) and $\{\zeta(23), \dots, \zeta(105)\}$ ($r=21$), so that “one of $\zeta(r+2), \dots, \zeta(5r)$ ” holds for every odd $r \leq 25$ (numerically for $r=17, 19, 23, 25$). We also determine the limit $r \rightarrow \infty$ of the construction: the dominant saddle point collapses onto the real axis, the criterion becomes linear in r , and the optimal ratio is $q/r \rightarrow 1 + \lambda_* \approx 5.15$; in particular the “5r” statement is out of reach of this family for $r \gtrsim 27$. Finally we prove a theorem uniform in r : *for every odd $r \geq 3$, at least one of $\zeta(r+2), \zeta(r+4), \dots, \zeta(6r-1)$ is irrational* (Zudilin's uniform bound is $8r-1$). The proof uses one explicit family of direction vectors with seven distinct directions; its phase function is r times a fixed function plus an r -independent perturbation, so that all analytic inequalities reduce to finitely many inequalities independent of r , verified once, plus scalar inequalities checked for $51 \leq r \leq 19999$ individually and for $r \geq 20001$ by explicit asymptotic bounds; the cases $17 \leq r \leq 49$ are certified one by one.

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1. STATEMENTS

Throughout, $r \geq 3$ and $q \geq r+4$ are odd integers, and $\boldsymbol{\eta} = (\eta_0; \eta_1, \dots, \eta_q)$ is a vector of positive integers (“directions”) satisfying

$$(1.1) \quad \eta_1 \leq \eta_2 \leq \dots \leq \eta_q < \tfrac{1}{2}\eta_0, \quad \sum_{j=1}^q \eta_j < \tfrac{1}{2}\eta_0 (q-r).$$

For $n \in \mathbb{N}$ put $h_0 = \eta_0 n + 2$, $h_j = \eta_j n + 1$ ($1 \leq j \leq q$) and define the rational function

$$(1.2) \quad R_n(t) = (h_0 + 2t) \prod_{j=1}^r \frac{(t+1)_{h_j-1} (t+h_0-h_j+1)_{h_j-1}}{((h_j-1)!)^2} \prod_{j=r+1}^q \frac{(h_0-2h_j)!}{(t+h_j)_{h_0-2h_j+1}},$$

where $(y)_m = y(y+1) \cdots (y+m-1)$, and the numbers

$$(1.3) \quad F_n = F_n(\boldsymbol{\eta}) = \frac{1}{(r-1)!} \sum_{t=0}^{\infty} R_n^{(r-1)}(t).$$

(This is $F(\mathbf{h})$ of [13, (8.6)–(8.7)]: R_n is the product of Zudilin’s elementary “bricks”, the first r factors are his $\Gamma(h_0+t)\Gamma(h_j+t)/(\Gamma(1+t)\Gamma(1+h_0-h_j+t))$ normalised by $(h_j-1)!^{-2}$.) The first r factors of (1.2) are polynomials in t (the “zero blocks”), the last $q-r$ are the “pole blocks”; the linear form F_n is obtained by differentiating $r-1$ times.

Theorem 1.1 ($r = 5$). *At least one of the eight numbers $\zeta(7), \zeta(9), \zeta(11), \dots, \zeta(21)$ is irrational.*

Theorem 1.2 ($r = 7$). *At least one of the thirteen numbers $\zeta(9), \zeta(11), \zeta(13), \dots, \zeta(33)$ is irrational.*

Theorem 1.3 ($r = 3$; Zudilin [10, 13]). *At least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational.*

Further instances ($r = 9, 11, 13, 15, 21$: one of $\zeta(11), \dots, \zeta(45)$; of $\zeta(13), \dots, \zeta(55)$; of $\zeta(15), \dots, \zeta(65)$; of $\zeta(17), \dots, \zeta(75)$; of $\zeta(23), \dots, \zeta(105)$), and the behaviour of the construction as $r \rightarrow \infty$, are in Section 7; the uniform statement “one of $\zeta(r+2), \dots, \zeta(6r-1)$ for every odd r ” is Theorem 8.1 in Section 8.

The direction vectors are

$$(A) \quad r = 5, \quad q = 23, \quad \eta_0 = 196, \\ \boldsymbol{\eta}' = (58, 58, 58, 59, 60; 62, 62, 63, 64, 64, 65, 66, 66, 68, 69, 70, 72, 73, 75, 76, 77, 79, 80); \quad (1.4)$$

- (B) $r = 7$, $q = 35$, $\eta_0 = 151$,
 $\boldsymbol{\eta}' = (46, 46, 46, 46, 46, 46, 46; 47, 47, 48, 49, 49, 49, 50, 51, 51, 51, 52, 53, 53, 53, 54,$
 $55, 55, 55, 56, 57, 57, 57, 58, 59, 59, 59, 60, 61); (1.5)$
(C) $r = 3$, $q = 13$, $\eta_0 = 90$, $\boldsymbol{\eta}' = (27, 27, 27; 28, 29, 30, 31, 32, 33, 34, 35, 36, 37); (1.6)$

where the semicolon separates the r zero-block directions from the $q - r$ pole-block directions. Condition (1.1) holds: $\sum \eta_j = 1544 < 1764$, $1827 < 2114$, $406 < 450$ respectively.

The proof has the usual three parts. (i) F_n is a rational linear form in $1, \zeta(r+2), \zeta(r+4), \dots, \zeta(q-2)$ (Lemma 2.2). (ii) An explicit integer Δ_n with $\Delta_n F_n \in \mathbb{Z} + \mathbb{Z}\zeta(r+2) + \dots + \mathbb{Z}\zeta(q-2)$ and $\lim \frac{1}{n} \log \Delta_n = C_2$ (Section 3; this is Zudilin's Lemma 19 plus the Chudnovsky–Rukhadze–Hata asymptotics). (iii) $\limsup \frac{1}{n} \log |F_n| \leq -C_0$ and $F_n \neq 0$ infinitely often (Section 4). If all of $\zeta(r+2), \dots, \zeta(q-2)$ were rational with common denominator A , then $A\Delta_n F_n$ would be a non-zero integer for infinitely many n while $|A\Delta_n F_n| \leq e^{n(C_2 - C_0 + o(1))} \rightarrow 0$ as soon as $C_0 > C_2$. The certified values are

	C_0	C_2	$C_0 - C_2$
(A)	939.0301751743	936.8905456295	+2.1396
(B)	1175.7847344814	1171.2347981360	+4.5499
(C)	230.8974422767	227.1012619982	+3.7962

(Section 5; for comparison, Zudilin's own parameters for Theorem 1.3 give $C_0 = 227.5802$, $C_2 = 226.2494$ [13, p. 285], which our code reproduces to all printed digits.)

2. THE LINEAR FORMS

Lemma 2.1. (a) $R_n(t) = O(t^{-2})$ as $|t| \rightarrow \infty$; more precisely $\deg R_n \leq 1 - q + r - n \leq -2$. (b) $R_n(-t - h_0) = -R_n(t)$. (c) The poles of R_n are the points $t = -k$, $h_{r+1} \leq k \leq h_0 - h_{r+1}$, of order at most $q - r$; R_n has a zero of order exactly r at each $t = -m$, $1 \leq m \leq h_1 - 1$, and at each $t = -m$, $h_0 - h_1 + 1 \leq m \leq h_0 - 1$.

Proof. (a) The numerator has degree $1 + 2 \sum_{j \leq r} (h_j - 1)$ and the denominator has degree $\sum_{j > r} (h_0 - 2h_j + 1)$, so $\deg R_n = 1 - q - r + 2 \sum_{j=1}^q h_j - (q - r)h_0$. With $h_j = \eta_j n + 1$, $h_0 = \eta_0 n + 2$ this is $1 - q + r + 2n(\sum_{j=1}^q \eta_j - \frac{1}{2}(q - r)\eta_0)$. By (1.1) the bracket is a negative half-integer, so $2n(\dots) \leq -n$ and $\deg R_n \leq 1 - q + r - n \leq -2$ because $q \geq r + 4$. (b) Under $t \mapsto -t - h_0$ the factor $h_0 + 2t$ changes sign; $(t + 1)_{h_j-1} \mapsto (-1)^{h_j-1} (t + h_0 - h_j + 1)_{h_j-1}$ and vice versa, so each zero block is multiplied by $(-1)^{2(h_j-1)} = 1$; each pole block is multiplied by $(-1)^{h_0-2h_j+1} = (-1)^{h_0+1}$. Since $q - r$ is even, the total sign is -1 . (c) The zeros of the j -th zero block are $-1, \dots, -(h_j - 1)$ and $-(h_0 - h_j + 1), \dots, -(h_0 - 1)$; the poles of the j -th pole block are $-h_j, \dots, -(h_0 - h_j)$. Since $h_1 \leq \dots \leq h_q < h_0/2$ these three families are pairwise disjoint from one another in the sense needed: for $1 \leq m \leq h_1 - 1$ all r zero blocks vanish simply at $-m$ and no pole block has a pole there ($h_j \geq h_{r+1} > h_1 - 1$), so the order is exactly r ; similarly at the mirror points. $\square$

By Lemma 2.1(c), $R_n^{(r-1)}(t) = 0$ for $1 - h_1 \leq t \leq -1$, so the sum (1.3) may be started at $t = 1 - h_1$; this is the form in which the arithmetic is analysed. Write the partial fraction expansion

$$(2.1) \quad R_n(t) = \sum_{i=1}^{q-r} \sum_{k=h_{r+1}}^{h_0-h_{r+1}} \frac{a_{i,k}}{(t+k)^i}, \quad a_{i,k} = \frac{1}{(q-r-i)!} \left(\frac{d}{dt} \right)^{q-r-i} \left[R_n(t)(t+k)^{q-r} \right]_{t=-k}.$$

Lemma 2.2. $a_{i,k} = (-1)^{i+1} a_{i,h_0-k}$; $\sum_k a_{1,k} = 0$; and

$$(2.2) \quad \begin{aligned} F_n &= \sum_{\substack{r+2 \leq s \leq q-2 \\ s \text{ odd}}} A_{s,n} \zeta(s) - A_{0,n}, \quad A_{s,n} = \binom{s-1}{r-1} \sum_k a_{s-r+1,k}, \\ A_{0,n} &= \sum_{i=1}^{q-r} \binom{i+r-2}{r-1} \sum_k a_{i,k} \sum_{\ell=1}^{k-h_1} \frac{1}{\ell^{i+r-1}}. \end{aligned}$$

In particular $F_n \in \mathbb{Q} + \mathbb{Q}\zeta(r+2) + \mathbb{Q}\zeta(r+4) + \cdots + \mathbb{Q}\zeta(q-2)$.

Proof. Differentiating (2.1) $r-1$ times and dividing by $(r-1)!$ gives

$$\frac{R_n^{(r-1)}(t)}{(r-1)!} = (-1)^{r-1} \sum_{i,k} \binom{i+r-2}{r-1} \frac{a_{i,k}}{(t+k)^{i+r-1}},$$

and $(-1)^{r-1} = 1$. Summing over $t \geq 1 - h_1$ and using $\sum_{t \geq 1-h_1} (t+k)^{-m} = \zeta(m) - \sum_{\ell=1}^{k-h_1} \ell^{-m}$ for $m \geq 2$ (note $k \geq h_{r+1} > h_1$, so $t+k \geq 1$), while for $m = 1$ (i.e. $i = 1, r = 1$ — which does not occur since $r \geq 3$) nothing is needed, we get (2.2) with all i from 1 to $q-r$. The symmetry $a_{i,k} = (-1)^{i+1} a_{i,h_0-k}$ follows from Lemma 2.1(b) and uniqueness of (2.1); hence $\sum_k a_{i,k} = 0$ for even i , i.e. $A_{s,n} = 0$ for even $s = i+r-1$. Finally $\sum_k a_{1,k}$ is the sum of all residues of R_n , which vanishes because $R_n = O(t^{-2})$; this kills $s = r$. The surviving odd $s = i+r-1$ with $3 \leq i \leq q-r$ odd are $r+2, r+4, \dots, q-2$. $\square$

3. ARITHMETIC OF THE COEFFICIENTS

Let $D_N = \text{lcm}(1, 2, \dots, N)$. Put

$$(3.1) \quad \hat{m}_0 = \max\{\eta_r, \eta_0 - 2\eta_{r+1}\}, \quad \hat{m}_j = \max\{\hat{m}_0, \eta_0 - \eta_1 - \eta_{r+j}\} \quad (1 \leq j \leq q-r),$$

so that $\hat{m}_1 \geq \hat{m}_2 \geq \cdots \geq \hat{m}_{q-r} \geq \hat{m}_0$. (In Zudilin's notation $m_0 = \max\{h_r - 1, h_0 - 2h_{r+1}\} = \hat{m}_0 n$ and $m_j = \max\{m_0, h_0 - h_1 - h_{r+j}\} = \hat{m}_j n$ exactly, because $h_0 - h_1 - h_{r+j} = (\eta_0 - \eta_1 - \eta_{r+j})n$.)

3.1. The function φ . For real x, y let

$$(3.2) \quad \begin{aligned} \varphi_0(x, y) &= \sum_{j=1}^r \left(\lfloor y \rfloor + \lfloor \eta_0 x - y \rfloor - \lfloor y - \eta_j x \rfloor - \lfloor (\eta_0 - \eta_j)x - y \rfloor - 2\lfloor \eta_j x \rfloor \right) \\ &\quad + \sum_{j=r+1}^q \left(\lfloor (\eta_0 - 2\eta_j)x \rfloor - \lfloor y - \eta_j x \rfloor - \lfloor (\eta_0 - \eta_j)x - y \rfloor \right), \end{aligned}$$

and $\varphi(x) = \min_{y \in \mathbb{R}} \varphi_0(x, y)$.

Lemma 3.1. φ_0 is 1-periodic in each variable, integer valued and ≥ 0 ; hence φ is 1-periodic, integer valued, ≥ 0 , and the minimum is attained. Moreover φ is constant on each open interval between consecutive points of the finite set

$$X = \{a/b \in [0, 1] : a \in \mathbb{Z}, b \in Q\}, \quad Q = \{|a_i b_{i'} - a_{i'} b_i| \neq 0\} \cup \{|a_i| : b_i = 0\} \cup \{\hat{m}_{q-r}\},$$

where (a_i, b_i) runs over the coefficient pairs of the linear forms $a_i x + b_i y$ appearing in (3.2), i.e. $(0, 1), (\eta_0, -1), (-\eta_j, 1), (\eta_0 - \eta_j, -1), (\eta_j, 0)$ for $j \leq r$ and $(\eta_0 - 2\eta_j, 0), (-\eta_j, 1), (\eta_0 - \eta_j, -1)$ for $j > r$.

Proof. Periodicity in y : the coefficients of y in the four y -dependent terms of each zero block sum to $1 - 1 - 1 + 1 = 0$ and in each pole block to $-1 + 1 = 0$; in x : $\eta_0 - (-\eta_j) \cdot 1 \dots$, explicitly each zero block contributes $\eta_0 + \eta_j - (\eta_0 - \eta_j) - 2\eta_j = 0$ and each pole block $(\eta_0 - 2\eta_j) + \eta_j - (\eta_0 - \eta_j) = 0$. Non-negativity: each pole-block summand is $\lfloor A + B \rfloor - \lfloor A \rfloor - \lfloor B \rfloor \in \{0, 1\}$ with $A = y - \eta_j x$, $B = (\eta_0 - \eta_j)x - y$; each zero-block summand is the sum of two such expressions, with $(A, B) =$

$(y - \eta_j x, \eta_j x)$ and $((\eta_0 - \eta_j)x - y, \eta_j x)$. For fixed x the function $y \mapsto \varphi_0(x, y)$ is piecewise constant with jumps only on the lines $a_i x + b_i y \in \mathbb{Z}$, $b_i \neq 0$; as x varies, the value of each $\lfloor a_i x + b_i y \rfloor$ on a cell of the arrangement of lines $\{a_i x + b_i y = k\}$ is constant, so the collection of values of $\varphi_0(x, \cdot)$ can only change at abscissae where two lines meet, $x = (kb_{i'} - k'b_i)/(a_i b_{i'} - a_{i'} b_i)$, or where a vertical line $a_i x = k$ ($b_i = 0$) is crossed. These abscissae lie in X . $\square$

For a prime p and $n \geq 1$ put $\Phi_n = \prod_{\sqrt{h_0} < p \leq \hat{m}_{q-r} n} p^{\varphi(n/p)}$ and

$$(3.3) \quad \Delta_n = D_{\hat{m}_1 n}^r \prod_{j=2}^{q-r} D_{\hat{m}_j n} \cdot \Phi_n^{-1}.$$

Lemma 3.2 (Zudilin [13, Lemma 19]). *For all $n \geq n_0(\eta)$, Δ_n is a positive integer and $\Delta_n A_{s,n} \in \mathbb{Z}$ for $s = 0$ and for all odd s with $r + 2 \leq s \leq q - 2$. Consequently $\Delta_n F_n \in \mathbb{Z} + \mathbb{Z}\zeta(r + 2) + \mathbb{Z}\zeta(r + 4) + \cdots + \mathbb{Z}\zeta(q - 2)$.*

Proof. This is exactly [13, Lemma 19] (whose proof is written for arbitrary odd r , $q \geq r + 4$ and directions satisfying (1.1)), with the minor difference that Zudilin's Φ has exponents $\nu_p = \min_{h_{r+1} \leq k \leq h_0 - h_{r+1}} \nu_{k,p}$ where $\nu_{k,p}$ is a sum of floor functions of the quantities $(k-1)/p$, $(h_0 - k - 1)/p$, $(k - h_j)/p$, $(h_0 - h_j - k)/p$, $(h_j - 1)/p$, $(h_0 - 2h_j)/p$; with $h_0 = \eta_0 n + 2$, $h_j = \eta_j n + 1$ one checks term by term that $\nu_{k,p} = \varphi_0(n/p, (k-1)/p)$, so $\nu_p \geq \varphi(n/p) \geq 0$ and $\Phi_n \mid \Phi$; dividing by the smaller integer Φ_n instead of Φ preserves the conclusion. For the reader's convenience we recall the argument. Write $R_n(t)(t+k)^{q-r}$ as the product of $h_0 + 2t$, the $2r$ integer-valued polynomial bricks $P_j^\pm(t) = (t+1)_{h_j-1}/(h_j-1)!$, $(t+h_0-h_j+1)_{h_j-1}/(h_j-1)!$ and the $q-r$ factors $C_j(t)(t+k)$. By the Leibniz rule, $a_{i,k}$ in (2.1) is a sum of products $\prod_\beta \frac{1}{l_\beta!} (\text{brick})^{(l_\beta)} (-k)$ over all distributions $\sum_\beta l_\beta = q - r - i$ of the derivatives among the bricks. By [13, Lemmas 15, 16], $D_{\hat{m}_0 n}^l \frac{1}{l!} (P_j^\pm)^{(l)}(-k) \in \mathbb{Z}$ and $D_{\hat{m}_0 n}^l \frac{1}{l!} (C_j(t)(t+k))^{(l)}|_{t=-k} \in \mathbb{Z}$ for every $l \geq 0$ and every $k \in [h_{r+1}, h_0 - h_{r+1}]$ (the lengths of the bricks are $h_j - 1 \leq \hat{m}_0 n$ and $h_0 - 2h_j \leq \hat{m}_0 n$), whence

$$(3.4) \quad D_{\hat{m}_0 n}^{q-r-i} a_{i,k} \in \mathbb{Z}.$$

By [13, Lemmas 17, 18] (applied with $p > \sqrt{h_0}$, so that $\text{ord}_p N! = \lfloor N/p \rfloor$ for all $N \leq h_0$ and $p \nmid l!$ for $l \leq q$ once $n \geq n_0$), the p -adic order of each brick derivative $\frac{1}{l!} (\text{brick})^{(l)}(-k)$ is at least $-l$ plus the corresponding summand of $\varphi_0(n/p, (k-1)/p)$; summing over the bricks and taking the minimum over the Leibniz terms,

$$(3.5) \quad \text{ord}_p a_{i,k} \geq -(q - r - i) + \varphi_0(n/p, (k-1)/p) \geq -(q - r - i) + \varphi(n/p) \quad (p > \sqrt{h_0}).$$

The coefficients $A_{s,n}$, $s \geq r + 2$. Here $A_{s,n} = \binom{s-1}{r-1} \sum_k a_{i,k}$ with $i = s - r + 1 \geq 3$, so by (3.4) already $D_{\hat{m}_0 n}^{q-r-i} A_{s,n} \in \mathbb{Z}$, and $D_{\hat{m}_0 n}^{q-r-i}$ divides the product of D 's in (3.3).

The constant term $A_{0,n}$. Consider the term $a_{i,k} \sum_{\ell \leq k-h_1} \ell^{-(i+r-1)}$ of (2.2). If $a_{i,k} \neq 0$ then $-k$ is a pole of R_n of order $\geq i$, i.e. k lies in the ranges $[h_j, h_0 - h_j]$ of at least i pole blocks; these ranges are nested and decreasing in j , hence $k \leq h_0 - h_{r+i}$ and $k - h_1 \leq h_0 - h_1 - h_{r+i} = (\eta_0 - \eta_1 - \eta_{r+i})n \leq \hat{m}_i n$. Thus the term has denominator dividing $D_{\hat{m}_i n}^{i+r-1} D_{\hat{m}_0 n}^{q-r-i}$. The product $D_{\hat{m}_1 n}^r D_{\hat{m}_2 n} \cdots D_{\hat{m}_{q-r} n}$ has $q - 1$ factors; its $i + r - 1$ largest factors $D_{\hat{m}_1 n}^r D_{\hat{m}_2 n} \cdots D_{\hat{m}_i n}$ are all divisible by $D_{\hat{m}_i n}$ and the remaining $q - r - i$ factors are divisible by $D_{\hat{m}_0 n}$. Hence the product of D 's in (3.3) times $A_{0,n}$ is an integer.

Removing Φ_n . Let $\sqrt{h_0} < p \leq \hat{m}_{q-r} n$. Since $\hat{m}_1 n < \eta_0 n < h_0 < p^2$, each of the $q - 1$ factors $D_{\hat{m}_j n}$ is divisible by p exactly once, and every $\ell \leq \hat{m}_1 n$ has $\text{ord}_p \ell \leq 1$. By (3.5), $\text{ord}_p A_{s,n} \geq -(q - r - i) + \varphi(n/p)$ and $\text{ord}_p A_{0,n} \geq \min_i (-(q - r - i) + \varphi(n/p) - (i + r - 1)) = -(q - 1) + \varphi(n/p)$. The p -part of Δ_n is $p^{q-1-\varphi(n/p)}$, so $\text{ord}_p(\Delta_n A_{s,n}) \geq 0$ for all s . For the other primes the p -part of

Δ_n equals that of the product of the D 's, and the previous two paragraphs apply. Finally $\Delta_n \in \mathbb{Z}$ because $\varphi \leq q-1$ (checked in §3.3: $\max \varphi = 18, 27, 11$ against $q-1 = 22, 34, 12$). $\square$

3.2. Asymptotics of Δ_n .

Lemma 3.3. $\lim_{n \rightarrow \infty} \frac{\log \Delta_n}{n} = C_2 := r\hat{m}_1 + \hat{m}_2 + \cdots + \hat{m}_{q-r} - I_\Phi$, where

$$(3.6) \quad I_\Phi = \int_0^1 \varphi(x) \psi'(1+x) dx + \int_{1/\hat{m}_{q-r}}^1 \frac{\varphi(x)}{x^2} dx = \int_0^1 \varphi(x) d\psi(x) - \int_0^{1/\hat{m}_{q-r}} \frac{\varphi(x)}{x^2} dx,$$

$\psi = \Gamma'/\Gamma$. If $0 = x_0 < x_1 < \cdots < x_N = 1$ are the points of X and $\varphi = \varphi_i$ on (x_{i-1}, x_i) , then

$$(3.7) \quad I_\Phi = \sum_{i=1}^N \varphi_i(\psi(1+x_i) - \psi(1+x_{i-1})) + \sum_{i: x_{i-1} \geq 1/\hat{m}_{q-r}} \varphi_i\left(\frac{1}{x_{i-1}} - \frac{1}{x_i}\right).$$

Proof. $\log D_N = N(1+o(1))$ is the prime number theorem. For Φ_n we use the lemma of Chudnovsky–Rukhadze–Hata in the form of [12, Lemma 4.4] (see also [4, Lemma 3.2]): for $0 < u < v < 1$ and $c > 0$, $\sum_{p > \sqrt{cn}, \{n/p\} \in [u,v]} \log p = n(\psi(v) - \psi(u)) + o(n)$. The primes $p > n$ with $n/p \in [u, v]$ contribute $n(1/u - 1/v) + o(n)$ by the prime number theorem, so the primes $p \leq n$ with $\{n/p\} \in [u, v]$ contribute $n \int_u^v \psi'(1+x) dx + o(n)$ (recall $\psi'(x) = \psi'(1+x) + x^{-2}$), and the primes $n < p \leq \hat{m}_{q-r}n$ with $n/p \in [u, v]$ contribute $n \int_{[u,v] \cap [1/\hat{m}_{q-r}, 1]} x^{-2} dx + o(n)$. Apply this to the finitely many intervals (x_{i-1}, x_i) on which φ is constant (Lemma 3.1; $1/\hat{m}_{q-r} \in X$); the primes with $n/p \in X$ are at most $|X|$ in number for each n and contribute $O(\log n)$. Since $\varphi \geq 0$ is bounded, the excluded primes $p \leq \sqrt{h_0}$ contribute $O(\sqrt{n} \log n)$. Formula (3.7) is (3.6) evaluated on the partition. $\square$

3.3. Numerical values. For the three direction vectors (1.4)–(1.6) the quantities in Lemma 3.3 are as follows ($\hat{m} = (\hat{m}_1, \dots, \hat{m}_{q-r})$; $|X|$ is the number of breakpoints; $\max \varphi$ the largest value of φ on $[0, 1)$):

	$\hat{m}$	$r\hat{m}_1 + \sum_{j \geq 2} \hat{m}_j$	$ X $	$\max \varphi$	I_Φ	C_2
(A)	(76, 76, 75, 74, 74, 73, $72^{\times 12}$)	1616	3461	18	679.1094543705	936.8905456295
(B)	(58, 58, $57^{\times 26}$)	1946	2231	27	774.7652018640	1171.2347981360
(C)	(35, $34^{\times 9}$)	411	879	11	183.8987380018	227.1012619982

How these were computed and how accurate they are. The set X was generated exactly (all denominators in Q are $\leq 2\eta_0 + \hat{m}_{q-r}$). On each interval, $\varphi_i = \min_y \varphi_0(\hat{x}, y)$ at the midpoint $\hat{x}$ was computed twice: once in floating point and once in exact rational arithmetic (the minimum over y is a minimum over the finitely many cells of $y \mapsto \varphi_0(\hat{x}, y)$, whose endpoints are the fractional parts of $-a_i \hat{x}/b_i$); the two computations agree on every interval. Thus the φ_i are exact integers. The values $\psi(1+x_i)$ were evaluated to 30 significant digits (`mpmath`); the sums (3.7) were then accumulated in double precision. With at most 3461 terms, each of absolute value $\leq 27 \cdot \max |\psi'| \leq 27 \cdot \pi^2/6$, the accumulated rounding error is below 10^{-9} , while the margins $C_0 - C_2$ in (1.7) exceed 2. (Zudilin's parameters for Theorem 1.3, $\eta_0 = 91$, $\eta_j = 27, 27, 27, 29, 30, \dots, 38$, give with the same code $I_\Phi = 176.75055734$ and $C_2 = 226.24944266$, in agreement with [13, p. 285] to all published digits.)

4. ANALYTIC ESTIMATES

4.1. Notation. We scale $t = nu$. Let $\Omega = \mathbb{C} \setminus ((-\infty, -\eta_1] \cup [0, \infty))$ and, with principal branches of the logarithm,

$$(4.1) \quad f(u) = r \left[(-u) \log(-u) + (\eta_0 + u) \log(\eta_0 + u) \right] + \sum_{j=1}^q \left[(\eta_j + u) \log(\eta_j + u) - (\eta_0 - \eta_j + u) \log(\eta_0 - \eta_j + u) \right] + \kappa_0,$$

$\kappa_0 = \sum_{j>r}(\eta_0 - 2\eta_j)\log(\eta_0 - 2\eta_j) - 2\sum_{j\leq r}\eta_j\log\eta_j$. Then f is analytic on Ω , real on $(-\eta_1, 0)$, $f(\bar{u}) = \overline{f(u)}$, and

$$(4.2) \quad f'(u) = r[\log(\eta_0 + u) - \log(-u)] + \sum_j [\log(\eta_j + u) - \log(\eta_0 - \eta_j + u)],$$

$$(4.3) \quad f''(u) = r\left[\frac{1}{\eta_0 + u} - \frac{1}{u}\right] + \sum_j \left[\frac{1}{\eta_j + u} - \frac{1}{\eta_0 - \eta_j + u}\right],$$

$$f'''(u) = -r\left[\frac{1}{(\eta_0 + u)^2} - \frac{1}{u^2}\right] - \sum_j \left[\frac{1}{(\eta_j + u)^2} - \frac{1}{(\eta_0 - \eta_j + u)^2}\right].$$

Put $G(u) = (\eta_0 + 2u)(-u)^{-r/2}(\eta_0 + u)^{3r/2} \prod_{j=1}^q (\eta_j + u)^{1/2}(\eta_0 - \eta_j + u)^{-3/2}$ (principal powers; analytic and non-vanishing on $\Omega \setminus \{-\eta_0/2\}$), and for $k \in \mathbb{Z}$

$$F_k(u) = f(u) + ik\pi u, \quad H_k(u) = \operatorname{Re} F_k(u) = \operatorname{Re} f(u) - k\pi \operatorname{Im} u.$$

4.2. Integral representation and Fourier modes. Let $\cot_r(z) = \frac{(-1)^{r-1}}{(r-1)!} \frac{d^{r-1}}{dz^{r-1}} \cot z$. By [12, Lemmas 2.2, 2.3], $\cot_r z = U_r(\cot z)$ for a polynomial U_r of degree r , and $\pi^r \cot_r(\pi t) = (t - m)^{-r} + O(1)$ near every $m \in \mathbb{Z}$; in particular $\cot_r(\pi t)$ is bounded on $\{|\operatorname{Im} t| \geq 1\}$ and on the lines $\operatorname{Re} t \in \frac{1}{2} + \mathbb{Z}$.

Lemma 4.1. *Let $M \in \mathbb{R}$ with $-h_1 < M < 0$. Then*

$$(4.4) \quad F_n = -\frac{1}{2\pi i} \int_{M-i\infty}^{M+i\infty} \pi^r \cot_r(\pi t) R_n(t) dt.$$

Proof. On the closed half plane $\operatorname{Re} t \geq M$ the function $\pi^r \cot_r(\pi t) R_n(t)$ is meromorphic with poles exactly at $t = 0, 1, 2, \dots$ (at the integers $-1, \dots, -(h_1 - 1)$ the pole of $\cot_r$ of order r is cancelled by the zero of R_n of order r , Lemma 2.1(c); the poles of R_n lie in $\operatorname{Re} t \leq -h_{r+1} < M$), and $\operatorname{Res}_{t=m} = R_n^{(r-1)}(m)/(r-1)!$ by the expansion of $\pi^r \cot_r(\pi t)$ at m . Integrate over the boundary of the rectangle with vertices $M \pm iN, N' + \frac{1}{2} \pm iN$: on the three sides other than $\operatorname{Re} t = M$ the factor $\cot_r(\pi t)$ is bounded and $R_n(t) = O(|t|^{-2})$, so these sides contribute $O(N^{-1} + N'^{-1})$. Letting $N, N' \rightarrow \infty$ and keeping track of the orientation (the side $\operatorname{Re} t = M$ is traversed downwards) gives (4.4). This is [12, Lemma 2.4] in our normalisation. $\square$

Lemma 4.2 (Eulerian expansion). *For odd $r \geq 3$,*

$$(4.5) \quad \sin^r(z) \cot_r(z) = \sum_{\substack{|k| \leq r-2 \\ k \equiv r \pmod{2}}} c_k e^{ikz}, \quad c_k = c_{-k} = \frac{A(r-1, \frac{r-2-k}{2})}{(r-1)!} > 0,$$

where $A(m, j)$ are the Eulerian numbers; in particular $c_{r-2} = 1/(r-1)!$. Explicitly $\sin^3 z \cot_3 z = \cos z$, $\sin^5 z \cot_5 z = \frac{1}{24}(e^{3iz} + 11e^{iz} + 11e^{-iz} + e^{-3iz})$, $\sin^7 z \cot_7 z = \frac{1}{720}(e^{5iz} + 57e^{3iz} + 302e^{iz} + 302e^{-iz} + 57e^{-3iz} + e^{-5iz})$.

Proof. Write $x = e^{2iz}$, so $\cot z = i + \frac{2i}{x-1}$ and $\frac{d}{dz} = 2ix \frac{d}{dx}$. The Eulerian identity $\sum_{\nu \geq 0} \nu^m x^\nu = \frac{x A_m(x)}{(1-x)^{m+1}}$, $A_m(x) = \sum_{j=0}^{m-1} A(m, j) x^j$ [3, (6.37)], gives $(x \frac{d}{dx})^m \frac{-1}{1-x} = -\frac{x A_m(x)}{(1-x)^{m+1}}$ and hence $\cot^{(m)}(z) = -2i(2i)^m \frac{x A_m(x)}{(1-x)^{m+1}}$ for $m \geq 1$. With $m = r-1$ and $\sin^r z = e^{-irz}(-1)^r(1-x)^r/(2i)^r$ one obtains $\sin^r z \cot_r z = \frac{1}{(r-1)!} e^{-irz} x A_{r-1}(x) = \frac{1}{(r-1)!} \sum_{j=0}^{r-2} A(r-1, j) e^{i(2j+2-r)z}$. The symmetry $A(m, j) = A(m, m-1-j)$ gives $c_k = c_{-k}$; $A(m, 0) = 1$ gives $c_{\pm(r-2)} = 1/(r-1)!$. The explicit cases are $A(2, \cdot) = (1, 1)$, $A(4, \cdot) = (1, 11, 11, 1)$, $A(6, \cdot) = (1, 57, 302, 302, 57, 1)$. $\square$

Next we write R_n through Gamma functions. Since $(t+1)_{h_j-1} = \Gamma(t+h_j)/\Gamma(t+1)$, $(t+h_0-h_j+1)_{h_j-1} = \Gamma(t+h_0)/\Gamma(t+h_0-h_j+1)$, $(t+h_j)_{h_0-2h_j+1} = \Gamma(t+h_0-h_j+1)/\Gamma(t+h_j)$ and $1/\Gamma(1+t) = -\Gamma(-t)\sin(\pi t)/\pi$,

$$(4.6) \quad R_n(t) = \left(-\frac{\sin \pi t}{\pi}\right)^r \mathcal{G}(t), \quad \mathcal{G}(t) := \text{Norm} \cdot (h_0+2t) \Gamma(-t)^r \Gamma(t+h_0)^r \prod_{j=1}^q \frac{\Gamma(t+h_j)}{\Gamma(t+h_0-h_j+1)},$$

$$\text{Norm} = \frac{\prod_{j>r}(h_0-2h_j)!}{\prod_{j\leq r}(h_j-1)!^2}.$$

$\mathcal{G}$ is meromorphic; in the half plane $\text{Re } t > -h_1$ its only poles are $t = 0, 1, 2, \dots$ (from $\Gamma(-t)^r$), and $\mathcal{G}(\bar{t}) = \overline{\mathcal{G}(t)}$. As r is odd, $\pi^r \cot_r(\pi t) R_n(t) = -\sin^r(\pi t) \cot_r(\pi t) \mathcal{G}(t)$, and Lemmas 4.1, 4.2 give

$$(4.7) \quad F_n = \frac{1}{2\pi i} \sum_k c_k I_{n,k}, \quad I_{n,k} := \int_{M-i\infty}^{M+i\infty} e^{ik\pi t} \mathcal{G}(t) dt, \quad \frac{I_{n,-k}}{2\pi i} = \overline{\left(\frac{I_{n,k}}{2\pi i}\right)},$$

the last identity because $\mathcal{G}(\bar{t}) = \overline{\mathcal{G}(t)}$ and the line is mapped to itself with reversed orientation by conjugation. Hence

$$(4.8) \quad F_n = 2 \sum_{k>0, k \equiv r(2)}^{r-2} c_k \text{Re} \left[\frac{1}{2\pi i} I_{n,k} \right].$$

4.3. Stirling's formula on admissible sets.

Lemma 4.3. *Let $K \subset \Omega$ be a set with $\inf_{u \in K} |u| > 0$, $\inf_{u \in K} \text{Re } u > -\eta_1$ and $\sup_{u \in K} |\arg(-u)| < \pi$. Then there are C_K and, for each $n \geq 1$, a function ε_n on K with $|\varepsilon_n(u)| \leq C_K/n$, such that*

$$(4.9) \quad \mathcal{G}(nu) = \mathcal{A}_n e^{nf(u)} G(u) (1 + \varepsilon_n(u)), \quad \mathcal{A}_n = c_0 n^{1-r-(q-r)/2}, \quad c_0 = (2\pi)^{\frac{q-r}{2}} \frac{\prod_{j>r}(\eta_0-2\eta_j)^{1/2}}{\prod_{j\leq r} \eta_j}.$$

Proof. With $t = nu$ the arguments of the Gamma functions in (4.6) are $-t = n(-u)$, $t+h_0 = n(\eta_0+u)+2$, $t+h_j = n(\eta_j+u)+1$, $t+h_0-h_j+1 = n(\eta_0-\eta_j+u)+2$, and $\text{Norm} = \prod_{j>r} \Gamma(n(\eta_0-2\eta_j)+1) / \prod_{j\leq r} \Gamma(n\eta_j+1)^2$. Using $\Gamma(z+1) = z\Gamma(z)$ we reduce all of them to $\Gamma(n\zeta)$ with $\zeta \in \{-u, \eta_0+u, \eta_j+u, \eta_0-\eta_j+u, \eta_0-2\eta_j, \eta_j\}$, at the price of the explicit polynomial factors $(v+1)v$, w , $(w'+1)w'$, etc. On K every such ζ satisfies $|\zeta| \geq c > 0$ and $|\arg \zeta| \leq \pi - \varepsilon_0$ (for $\zeta = -u$ by hypothesis; the others have positive real part because $\text{Re } u > -\eta_1$). By Stirling's formula in the form $\Gamma(z) = \sqrt{2\pi} z^{z-1/2} e^{-z} (1 + \rho(z))$ with $|\rho(z)| \leq C_{\varepsilon_0}/|z|$ uniformly in $|\arg z| \leq \pi - \varepsilon_0$ [9, §12.33], [6, Ch. 8, §4], applied with $z = n\zeta$ and principal logarithms ($\log(n\zeta) = \log n + \log \zeta$), the exponent $n \sum \pm \zeta \log \zeta$ is exactly $nf(u)$ (the sum of the coefficients of ζ over all Gamma factors, counted with multiplicity, is 0, so the terms $n\zeta \log n$ and $-n\zeta$ cancel and κ_0 is contributed by Norm). Collecting the factors $\sqrt{2\pi}$, the powers $(n\zeta)^{-1/2}$ and the polynomial factors gives $\mathcal{A}_n$, $G(u)$ and a factor $(1 + O(1/n))$; a direct count yields the exponent $1-r-(q-r)/2$ of n and the constant c_0 . $\square$

Lemma 4.4 (decay). *Fix n and $\varrho > 0$. There are C, c (depending on n, ϱ) such that for $|\text{Re } t| \leq \varrho n$, $|\text{Im } t| \geq 1$ and $|k| \leq r-2$: $|e^{ik\pi t} \mathcal{G}(t)| \leq C(1+|t|)^c e^{-2\pi |\text{Im } t|}$.*

Proof. For x in a compact set and $|y| \geq 1$, $|\Gamma(x+iy)| \leq C|y|^{x-1/2} e^{-\pi|y|/2}$ [9, §12.33]. In (4.6) the factors $\Gamma(-t)^r \Gamma(t+h_0)^r$ contribute $e^{-r\pi |\text{Im } t|}$, the ratios $\Gamma(t+h_j)/\Gamma(t+h_0-h_j+1)$ contribute e^0 , and $|e^{ik\pi t}| = e^{-k\pi \text{Im } t} \leq e^{(r-2)\pi |\text{Im } t|}$. $\square$

4.4. A Laplace lemma.

Lemma 4.5. *Let $\mathcal{L}$ be a piecewise smooth path, F and G analytic on a neighbourhood of $\mathcal{L}$, and let $u_* \in \mathcal{L}$, $\phi \in \mathbb{R}$, $s_0, \delta > 0$ be such that*

- (i) $F'(u_*) = 0$, $F''(u_*) \neq 0$, $e^{2i\phi}F''(u_*) = -|F''(u_*)|$, $G(u_*) \neq 0$, and the segment $\sigma = \{u_* + se^{i\phi} : |s| \leq s_0\}$ is contained in $\mathcal{L}$ (traversed in the direction of increasing s);
- (ii) $\operatorname{Re} F(u_* + se^{i\phi}) \leq \operatorname{Re} F(u_*) - \frac{1}{4}|F''(u_*)|s^2$ for $|s| \leq s_0$;
- (iii) $\operatorname{Re} F(u) \leq \operatorname{Re} F(u_*) - \delta$ for $u \in \mathcal{L} \setminus \sigma$;
- (iv) $J := \int_{\mathcal{L}} |e^{F(u)}G(u)| |du| < \infty$.

Let ε_n be measurable on $\mathcal{L}$ with $|\varepsilon_n| \leq C/n$. Then, as $n \rightarrow \infty$,

$$\int_{\mathcal{L}} e^{nF(u)}G(u)(1 + \varepsilon_n(u)) du = e^{nF(u_*)} \left[e^{i\phi}G(u_*) \sqrt{\frac{2\pi}{n|F''(u_*)|}} + o(n^{-1/2}) \right].$$

Proof. On $\mathcal{L} \setminus \sigma$ we have $|e^{n(F-F(u_*))}| \leq e^{-(n-1)\delta}|e^{F-F(u_*)}|$ by (iii), so that part of the integral is $O(e^{-n\delta}J)$ relative to $e^{nF(u_*)}$. On σ put $u = u_* + se^{i\phi}$; then $F(u) - F(u_*) = -\frac{1}{2}|F''(u_*)|s^2 + \rho(s)$ with $|\rho(s)| \leq M_3|s|^3/6$, $M_3 = \max_{|w| \leq s_0} |F'''(u_* + w)|$, and $G(u) = G(u_*) + O(|s|)$. Let $s_n = n^{-2/5}$. For $s_n \leq |s| \leq s_0$, (ii) bounds the integrand by $e^{-n|F''(u_*)|s_n^2/4} \max_{\sigma} |G|(1 + C/n)$, which is $O(e^{-cn^{1/5}})$. For $|s| \leq s_n$: $n|\rho(s)| \leq M_3n^{-1/5}/6$, so $e^{n\rho} = 1 + O(n^{-1/5})$, $G = G(u_*)(1 + O(n^{-2/5}))$, $1 + \varepsilon_n = 1 + O(n^{-1})$, and $\int_{-s_n}^{s_n} e^{-n|F''(u_*)|s^2/2} ds = \sqrt{2\pi/(n|F''(u_*)|)} (1 + O(e^{-cn^{1/5}}))$. Multiplying by $du = e^{i\phi}ds$ gives the claim. $\square$

4.5. Deformation of the contour.

Lemma 4.6. *Let $x_0 \in (-\eta_1, 0)$ and let $\mathcal{L}$ be a piecewise smooth path from $-i\infty$ to $+i\infty$ which meets the real axis only at x_0 , satisfies $\inf_{\mathcal{L}} \operatorname{Re} u > -\eta_1$, and coincides with vertical rays $\{\operatorname{Re} u = \text{const}\}$ outside a compact set. Then for every n and $|k| \leq r-2$, with $M = nx_0$ in (4.7),*

$$I_{n,k} = n \int_{\mathcal{L}} e^{ik\pi nu} \mathcal{G}(nu) du.$$

Proof. The integrand $u \mapsto e^{ik\pi nu} \mathcal{G}(nu)$ is meromorphic on $\{\operatorname{Re} u > -\eta_1\}$ with poles only at $u \in \frac{1}{n}\mathbb{Z}_{\geq 0} \subset [0, \infty)$. The vertical line $\operatorname{Re} u = x_0$ and $\mathcal{L}$ both cross the real axis exactly once, at $x_0 < 0$; the region between them (in each half plane) is therefore contained in $\{\operatorname{Im} u \neq 0\} \cup \{x_0\}$ and contains no pole. Truncate at $|\operatorname{Im} u| = Y'$ and close with horizontal segments; by Lemma 4.4 these segments contribute $O(Y'^c e^{-2\pi n Y'}) \rightarrow 0$. Cauchy's theorem gives the claim. $\square$

4.6. The main analytic proposition.

Proposition 4.7. *Let $k_d = r-2$, $F = F_{k_d}$. Suppose there exist $x_0 \in (-\eta_1, 0)$, a path $\mathcal{L}$ as in Lemma 4.6, a point $u_* \in \mathcal{L}$ with $\operatorname{Im} u_* < 0$, and $\phi, s_0, \delta > 0$ such that*

- (P1) $F'(u_*) = 0$, i.e. $f'(u_*) = -ik_d\pi$; $f''(u_*) \neq 0$; $e^{2i\phi}f''(u_*) = -|f''(u_*)|$; $\sigma = \{u_* + se^{i\phi} : |s| \leq s_0\} \subset \mathcal{L} \cap \{\operatorname{Im} u < 0\}$;
- (P2) $H_{k_d}(u_* + se^{i\phi}) \leq H_{k_d}(u_*) - \frac{1}{4}|f''(u_*)|s^2$ for $|s| \leq s_0$;
- (P3) $H_{k_d}(u) \leq H_{k_d}(u_*) - \delta$ for $u \in \mathcal{L} \setminus \sigma$;
- (P4) $H_k(u) \leq H_{k_d}(u_*) - \delta$ for all $u \in \mathcal{L}$ and all $k \in \{1, 3, \dots, r-4\}$;
- (P5) $\mathcal{L}$ satisfies the hypotheses of Lemma 4.3 (i.e. $\inf_{\mathcal{L}} |u| > 0$, $\inf_{\mathcal{L}} \operatorname{Re} u > -\eta_1$, $\sup_{\mathcal{L}} |\arg(-u)| < \pi$).

Put $H_d = H_{k_d}(u_*) = \operatorname{Re} F(u_*)$, $\alpha = \operatorname{Im} F(u_*)$ and $B = e^{i\phi}G(u_*)\sqrt{2\pi/|f''(u_*)|} \neq 0$, $-iB = |B|e^{i\beta}$. Then

$$(4.10) \quad F_n = \frac{c_{r-2} \mathcal{A}_n |B|}{\pi} n^{1/2} e^{nH_d} \left[\cos(n\alpha + \beta) + o(1) \right] \quad (n \rightarrow \infty).$$

Consequently (a) $\limsup_{n \rightarrow \infty} \frac{1}{n} \log |F_n| \leq H_d$; (b) if $\alpha \notin \pi\mathbb{Z}$, then $F_n \neq 0$ for infinitely many n (indeed for at least one of any two consecutive sufficiently large n).

Proof. By Lemma 4.6 and Lemma 4.3 (applicable by (P5)), $I_{n,k} = n\mathcal{A}_n \int_{\mathcal{L}} e^{nF_k(u)} G(u) (1 + \varepsilon_n(u)) du$ for every k . For $k = k_d$, Lemma 4.5 applies with $F = F_{k_d}$ by (P1)–(P3) (its hypothesis (iv) follows from (P2), (P3) and the decay of $e^{F_k} G$ along the vertical rays, Lemma 4.4 applied with $n = 1$): $\int_{\mathcal{L}} = e^{nF(u_*)} [Bn^{-1/2} + o(n^{-1/2})]$. For $k \in \{1, 3, \dots, r-4\}$, (P4) gives $|e^{nF_k(u)}| \leq e^{(n-1)(H_d-\delta)} |e^{F_k(u)}|$ on $\mathcal{L}$, whence $\int_{\mathcal{L}} = O(e^{n(H_d-\delta)})$. Inserting into (4.8):

$$F_n = 2c_{r-2} \operatorname{Re} \left[\frac{n\mathcal{A}_n}{2\pi i} e^{nF(u_*)} (Bn^{-1/2} + o(n^{-1/2})) \right] + O(n\mathcal{A}_n e^{n(H_d-\delta)}),$$

which is (4.10) because $\operatorname{Re}[-ie^{in\alpha} B] = |B| \cos(n\alpha + \beta)$. (a) is immediate since $\log \mathcal{A}_n = O(\log n)$. For (b): if $|\cos x| < c$ and $|\cos(x + \alpha)| < c$ then, using $|\sin y| \geq \frac{2}{\pi} \operatorname{dist}(y, \pi\mathbb{Z})$, both $x - \frac{\pi}{2}$ and $x + \alpha - \frac{\pi}{2}$ lie within $\frac{\pi}{2}c$ of $\pi\mathbb{Z}$, so $\operatorname{dist}(\alpha, \pi\mathbb{Z}) \leq \pi c$. Taking $c = \frac{1}{2\pi} \operatorname{dist}(\alpha, \pi\mathbb{Z}) > 0$ we get $\max\{|\cos(n\alpha + \beta)|, |\cos((n+1)\alpha + \beta)|\} \geq c$ for every n , and (4.10) gives $F_n \neq 0$ for at least one of $n, n+1$ once n is large. $\square$

4.7. Two elementary bounds used in the verification.

Lemma 4.8 (segment bound). *Let $[a, b] \subset \overline{\Omega}$ be a segment meeting $\mathbb{R}$ at most in an endpoint lying in $(-\eta_1, 0)$, and let $d_\lambda = \operatorname{dist}([a, b], -\lambda)$ for $\lambda \in \{0, \eta_0\} \cup \{\eta_j, \eta_0 - \eta_j\}_{j \leq q}$ (all > 0). Put $B_2 = r(d_0^{-1} + d_{\eta_0}^{-1}) + \sum_j (d_{\eta_j}^{-1} + d_{\eta_0 - \eta_j}^{-1})$. For $N \geq 1$ let $u_\nu = a + \frac{\nu}{N}(b - a)$ and $h = |b - a|/N$. Then for every $k \geq 0$*

$$\sup_{u \in [a, b]} H_k(u) \leq \max_{0 \leq \nu \leq N} H_k(u_\nu) + \frac{h}{2} \left(\max_\nu |f'(u_\nu)| + \frac{h}{2} B_2 + k\pi \right).$$

Proof. By (4.3), $|f''| \leq B_2$ on $[a, b]$, so $|f'(u) - f'(u_\nu)| \leq B_2|u - u_\nu|$ and $\sup |f'| \leq \max_\nu |f'(u_\nu)| + hB_2/2$. Along the unit-speed parametrisation, $|\frac{d}{ds} H_k| = |\operatorname{Re}(e^{i\theta} F'_k(u))| \leq |f'(u)| + k\pi$, and every point of the segment is within $h/2$ of a grid point. $\square$

Lemma 4.9 (tail bound). *Let $x_0 > -\eta_1$, $k \geq 0$, $Y \geq 1$ and $u = x_0 + iy$ with $|y| \geq Y$. Then*

$$H_k(u) \leq \beta_\pm(|y|) := -(r \pm k)\pi|y| + c_2 \log |y| + K, \quad (+ \text{ if } y > 0, - \text{ if } y < 0),$$

with $c_2 = r(|x_0| + x_0) + 2 \sum_j \eta_j - (q - r)\eta_0$ and $K = \kappa_0 + r(\frac{x_0^2}{Y} + |x_0|) + r(\frac{(\eta_0 + x_0)^2}{Y} + \eta_0 + x_0) + \sum_j (\frac{(\eta_j + x_0)^2}{Y} + \eta_j + x_0)$. If $c_2 \leq 0$ then $\beta_\pm$ is decreasing on $[Y, \infty)$, so $\sup_{|y| \geq Y} H_k(x_0 + iy) \leq \beta_\pm(Y)$.

Proof. For $a + x_0 > 0$: $\operatorname{Re}[(a + u) \log(a + u)] = (a + x_0) \log |a + u| - y \arg(a + u)$ with $\arg(a + u) = \operatorname{sgn}(y) \arctan \frac{|y|}{a + x_0}$, and $\arctan z \geq \frac{\pi}{2} - \frac{1}{z}$ for $z > 0$; hence $\operatorname{Re}[(a + u) \log(a + u)] \leq (a + x_0) \log(a + x_0 + |y|) - \frac{\pi}{2}|y| + (a + x_0)$. For the term $(-u) \log(-u)$: if $x_0 < 0$ the same computation with $a = 0$ applies; if $x_0 \geq 0$ then $-u = -x_0 + i(-y)$ has $|\arg(-u)| \geq \frac{\pi}{2}$ and $\operatorname{Re}(-u) \log |u| \leq 0$, giving $\operatorname{Re}((-u) \log(-u)) \leq -\frac{\pi}{2}|y|$; in both cases $\operatorname{Re}((-u) \log(-u)) \leq |x_0| \log(|x_0| + |y|) - \frac{\pi}{2}|y| + |x_0|$. For the terms with a minus sign ($b = \eta_0 - \eta_j$, $b + x_0 > 0$): $-\operatorname{Re}[(b + u) \log(b + u)] = -(b + x_0) \log |b + u| + y \arg(b + u) \leq -(b + x_0) \log |y| + \frac{\pi}{2}|y|$. Finally $\operatorname{Re}(ik\pi u) = -k\pi y$ and $\log(a + x_0 + |y|) \leq \log |y| + (a + x_0)/Y$. Summing: the terms $\mp \frac{\pi}{2}|y|$ cancel within each j -pair and add up to $-r\pi|y|$ from the two r -fold terms; the logarithmic coefficients add up to c_2 ; the constants to K . Monotonicity: $\beta'_\pm(y) = -(r \pm k)\pi + c_2/y < 0$ when $c_2 \leq 0$. $\square$

5. THE VERIFIED DATA AND THE PROOFS OF THEOREMS 1.1, 1.2, 1.3

5.1. The paths. For each of the three direction vectors we use a path $\mathcal{L}$ of the following shape. Let u_* be the saddle point of F_{k_d} in the lower half plane listed in Table 1, $e = e^{i\phi}$ with ϕ as in (P1), $s_0 = s_1 = 1$, and

$$\text{top} = u_* + s_0 e, \quad \text{bot} = u_* - s_0 e, \quad \text{ext} = u_* - (s_0 + s_1) e.$$

Then $\mathcal{L}$ is, in the order from $-i\infty$ to $+i\infty$: the vertical ray $\{\operatorname{Re} u = \operatorname{Re} \text{ext}, \operatorname{Im} u \leq \operatorname{Im} \text{ext}\}$; the segment $[\text{ext}, \text{bot}]$; the segment $\sigma = [\text{bot}, \text{top}]$ through u_* ; the segment $[\text{top}, x_0]$; the vertical ray $\{\operatorname{Re} u = x_0, \operatorname{Im} u \geq 0\}$. All points of $\mathcal{L}$ other than x_0 have $\operatorname{Im} u \neq 0$ (Table 2: $\operatorname{Im} \text{top} < 0$, and the segments $[\text{ext}, \text{bot}]$, $[\text{top}, x_0]$ meet $\mathbb{R}$ only at x_0). The constants of Lemma 4.3 are read off from Table 2: $\inf_{\mathcal{L}} |u| = \operatorname{dist}(\mathcal{L}, 0) > 1.7$, $\inf_{\mathcal{L}} \operatorname{Re} u = x_0 > -\eta_1$, $\sup_{\mathcal{L}} |\arg(-u)| < 2.02 < \pi$. Thus (P5) and the hypotheses of Lemma 4.6 hold.

TABLE 1. Saddle points u_k of $F_k = f + ik\pi u$ in the lower half plane ($f'(u_k) = -ik\pi$), the values $H_k(u_k)$, $f''(u_k)$, and $\alpha_k = \operatorname{Im} F_k(u_k) \bmod \pi$. The dominant mode is $k_d = r - 2$; $C_0 = -H_{k_d}(u_{k_d})$. Newton residuals $|f'(u_k) + ik\pi| < 10^{-49}$ at 50 digits.

	k	u_k	$H_k(u_k)$	$f''(u_k)$	$\alpha_k \bmod \pi$
(A)	1	$-6.538442182449 - 3.387946568365i$	-980.968288736	$0.823837268 - 0.295203977i$	0.205197
	3	$-0.590758658259 - 9.722688491285i$	-939.030175174326	$0.219466732 - 0.473810380i$	1.773388
(B)	1	$-4.485629507089 - 1.642867103557i$	-1233.606836513	$1.791796334 - 0.483934337i$	1.157548
	3	$-2.563290879243 - 4.779835583940i$	-1213.240330004	$0.998222261 - 1.084898242i$	2.977616
	5	$2.414446160185 - 6.698929974253i$	-1175.784734481369	$0.003016599 - 0.866587262i$	0.357148
(C)	1	$-3.204559550730 - 3.103130294133i$	-230.897442276750	$0.748174102 - 0.426828129i$	0.718053

TABLE 2. Path data and certified bounds. $\theta = H_d - \delta$ with $\delta = |f''(u_*)|s_0^2/8$ is the threshold in (P3), (P4). “sup” entries are the right-hand sides of Lemma 4.8 (grid points: $N = 20000$ on segments, $N = 400000$ on the rays up to $|\operatorname{Im} u| = Y$), “tail” entries are $\beta_{\pm}(Y)$ of Lemma 4.9 (with x_0 for the upper ray and $\operatorname{Re} \text{ext}$ for the lower ray; $c_2 < 0$ in all cases).

	(A)	(B)	(C)
ϕ	2.139308100	2.354453992	1.830016624
$ f''(u_*) $	0.522170396	0.866592513	0.861363302
$M_3 = \sup_{ w \leq 1} f'''(u_* + w) \leq$	0.072404	0.202912	0.273693
$3 f''(u_*) /(2M_3) \ (\geq s_0 = 1)$	10.82	6.41	4.72
$\delta = f''(u_*) /8$	0.065271	0.108324	0.107670
$\theta = H_d - \delta$	-939.095446	-1175.893059	-231.005113
x_0	-2	-2	-3.460886528
top	$-1.129137170 - 8.879985354i$	$1.708571167 - 5.990593547i$	$-3.460886528 - 2.136540163i$
bot	$-0.052380146 - 10.565391629i$	$3.120321153 - 7.407266402i$	$-2.948232574 - 4.069720425i$
ext	$0.485998366 - 11.408094766i$	$3.826196146 - 8.115602829i$	$-2.691905597 - 5.036310556i$
Y	784	604	360
$\operatorname{dist}(\mathcal{L}, 0); \sup_{\mathcal{L}} \arg(-u)$	1.99; 1.614	1.70; 2.012	3.46; 1.564
sup H_{k_d} on $[\text{top}, x_0]$	-939.296536	-1176.237722	-231.345825
sup H_{k_d} on upper ray, $0 \leq \operatorname{Im} u \leq Y$	-970.476007	-1227.711185	-235.559133
sup H_{k_d} on $[\text{ext}, \text{bot}]$	-939.281972	-1176.194938	-231.308192
sup H_{k_d} on lower ray, $-Y \leq \operatorname{Im} u \leq \operatorname{Im} \text{ext}$	-939.979980	-1177.318472	-232.456263
tail $\beta_+(Y)$ (upper ray)	-18007.6	-21075.9	-4042.6
tail $\beta_-(Y)$ (lower ray)	-3119.3	-1443.7	-1768.0
sup H_1 on $\mathcal{L}$, $ \operatorname{Im} u \leq Y$; tails	$-967.655; -13081.6, -8045.3$	$-1220.734; -13485.8, -9033.8$	—
sup H_3 on $\mathcal{L}$, $ \operatorname{Im} u \leq Y$; tails	—	$-1205.811; -17280.9, -5238.7$	—

5.2. Verification of (P1)–(P5). (P1). u_* is the root of $f'(u) + ik_d\pi = 0$ in Table 1 (obtained from the exact integer polynomial $P(\tau) = (\tau - \eta_0)^r \prod_j (\tau - \eta_j) - \tau^r \prod_j (\tau - \eta_0 + \eta_j)$, whose roots are the solutions of $e^{f'(u)} = \pm 1$, $u = \tau - \eta_0$, followed by Newton iteration on $f'(u) + ik_d\pi$ itself; since $f''(u_*) \neq 0$ and the residual is $< 10^{-49}$, the Newton–Kantorovich theorem places an exact zero within 10^{-48} of the printed value, which does not affect any of the inequalities below). $f''(u_*) \neq 0$ by the table; $\phi = \frac{1}{2}(\pi - \arg f''(u_*))$ (plus π , so that $\operatorname{Im} e^{i\phi} > 0$), whence $e^{2i\phi} f''(u_*) = -|f''(u_*)|$. Since $s_0 = 1 < |\operatorname{Im} u_*|$, σ lies in the lower half plane.

(P2). For $|w| \leq s_0$ the disc $|u - u_*| \leq s_0$ lies in $\Omega \cap \{\operatorname{Im} u < 0\}$ and $|f'''| \leq M_3$ there, with M_3 bounded by (4.3) using the distances from u_* to the branch points minus s_0 (Table 2). Taylor's formula gives $F(u_* + se^{i\phi}) = F(u_*) - \frac{1}{2}|f''(u_*)|s^2 + \rho(s)$, $|\rho(s)| \leq M_3|s|^3/6 \leq \frac{1}{4}|f''(u_*)|s^2$ for $|s| \leq s_0 \leq 3|f''(u_*)|/(2M_3)$, which holds (row 4 of Table 2).

(P3). On $[\text{ext}, \text{bot}]$, $[\text{top}, x_0]$ and on the two rays truncated at $|\operatorname{Im} u| = Y$, Lemma 4.8 gives the certified suprema in Table 2, all $< \theta$. Beyond $|\operatorname{Im} u| = Y$, Lemma 4.9 applies (with x_0 , resp. $\operatorname{Re} \text{ext}$, both $> -\eta_1$, and $c_2 < 0$ because $r(|x_0| + x_0) < (q - r)\eta_0 - 2\sum_j \eta_j$, cf. (1.1)); the values $\beta_{\pm}(Y)$ are far below θ . Together with (P2) at the endpoints of σ (where $H_{k_d} \leq H_d - 2\delta$), this is (P3) with the stated δ .

(P4). The same checks, now for H_k with $k \in \{1, 3, \dots, r-4\}$ and including the segment σ itself, give the last rows of Table 2, all $< \theta$.

(P5). See §5.1.

The phase. $\alpha = \operatorname{Im} F_{k_d}(u_*)$ has $\operatorname{dist}(\alpha, \pi\mathbb{Z}) \geq 0.357$ in all three cases (Table 1), so $\alpha \notin \pi\mathbb{Z}$.

5.3. Proof of Theorems 1.1, 1.2, 1.3. Fix one of the direction vectors (1.4)–(1.6). By §5.2, Proposition 4.7 applies, so with $C_0 := -H_d$ (Table 1, equation (1.7)) we have $\limsup \frac{1}{n} \log |F_n| \leq -C_0$ and $F_n \neq 0$ for infinitely many n . By Lemmas 3.2 and 3.3, $\Delta_n F_n \in \mathbb{Z} + \sum_s \mathbb{Z}\zeta(s)$ (s odd, $r+2 \leq s \leq q-2$) and $\log \Delta_n = n(C_2 + o(1))$ with C_2 as in §3.3.

Suppose, for contradiction, that $\zeta(r+2), \zeta(r+4), \dots, \zeta(q-2)$ are all rational, and let $A \in \mathbb{N}$ be a common denominator. Then $N_n := A \Delta_n F_n \in \mathbb{Z}$ for $n \geq n_0$, and

$$\log |N_n| \leq \log A + n(C_2 - C_0 + o(1)) \rightarrow -\infty$$

because $C_0 - C_2 > 0$ (equal to 2.1396, 4.5499, 3.7962 respectively, (1.7)). Hence $N_n = 0$, i.e. $F_n = 0$, for all large n , contradicting Proposition 4.7(b). Therefore at least one of $\zeta(r+2), \dots, \zeta(q-2)$ is irrational: for (A) one of $\zeta(7), \dots, \zeta(21)$; for (B) one of $\zeta(9), \dots, \zeta(33)$; for (C) one of $\zeta(5), \dots, \zeta(11)$. $\square$

Remark 5.1. (i) For (B) the dominant saddle has $\operatorname{Re} u_* = 2.41 > 0$, i.e. $\operatorname{Re} \tau_0 = \eta_0 + \operatorname{Re} u_* > \eta_0$ in Zudilin's variable; this is the situation excluded by the hypothesis “ $\operatorname{Re} \tau_0 < \eta_0$ ” of [13, Lemma 20] (stated there for $r = 3$ only). The path of §5.1 handles it because the exact integrand has no singularities in the open lower half plane, so the contour may leave the strip $-\eta_1 < \operatorname{Re} u < 0$ there. (ii) The r zero blocks and the $r-1$ derivatives are tied together: (P1)–(P3) concern the mode $k = r-2$, and for larger k the saddle moves away from the real axis ($\operatorname{Im} u_* = -3.1, -9.7, -6.7$ for $r = 3, 5, 7$), where the zeros of R_n no longer make $|\mathcal{G}|$ small; this is why the length of the list of zeta values grows with r . (iii) With Zudilin's own directions for $r = 3$ ($\eta_0 = 91$, $\boldsymbol{\eta}' = (27, 27, 27; 29, \dots, 38)$) the same programme returns $u_* = -3.520994582 - 3.328206906i$, $C_0 = 227.5801964127$, $C_2 = 226.2494426639$, reproducing [13, p. 285] to all published digits; the directions (1.6) merely give a larger margin.

6. WHAT IS VERIFIED BY COMPUTER, AND HOW

Every non-classical ingredient of the proof is one of finitely many explicit inequalities between computable real numbers, listed in Tables 1, 2 and in §3.3. They were checked with the Python programs `zud.py` and `certify.py` (available with this note) as follows.

- (1) *Saddle points and C_0 .* $P(\tau)$ is built with exact integer coefficients; its roots are computed with `mpmath.polyroots` at 90 digits; the roots with $\operatorname{Im} \tau > 0$, $\operatorname{Re} \tau > \eta_0 - \eta_{r+1}$ are mapped to $u = \tau - \eta_0 - i \operatorname{Im} \tau$ and polished by Newton's method on $f'(u) + ik\pi$ at 50 digits (residual $< 10^{-49}$); k is read off as $-\operatorname{Im} f'(u)/\pi$. $H_k(u_k)$, $f''(u_k)$, α_k and $G(u_k)$ are evaluated at 50 digits from (4.1), (4.3) with principal branches. Since $F'_k(u_k) = 0$, an error ϵ in u_k changes $H_k(u_k)$ by $O(\epsilon^2)$; all printed digits are reliable.
- (2) M_3 . From (4.3), $|f'''(u)| \leq r(|\eta_0 + u|^{-2} + |u|^{-2}) + \sum_j (|\eta_j + u|^{-2} + |\eta_0 - \eta_j + u|^{-2})$, maximised over the disc $|u - u_*| \leq s_0$ by replacing each distance by its minimum over the disc.

- (3) *Grid checks* (Lemma 4.8). H_k and $|f'|$ are evaluated in double precision at the $N + 1$ grid points; B_2 is computed from the exact distances of the segment to the branch points; the reported “sup” is $\max_\nu H_k(u_\nu) + \frac{h}{2}(\max_\nu |f'(u_\nu)| + \frac{h}{2}B_2 + k\pi) + 10^{-9}$, the last term dominating the double-precision rounding error of the evaluations (the summands of H_k are of size $\leq 10^4$, so the rounding error is $< 10^{-10}$).
- (4) *Tails* (Lemma 4.9). c_2 , K and $\beta_\pm(Y)$ are elementary expressions in the data, evaluated in double precision; the slack is several hundred units.
- (5) *Arithmetic side*. The set X is generated exactly; the values φ_i are computed in exact rational arithmetic (and agree with an independent floating-point evaluation on every interval); $\psi(1+x_i)$ is computed to 30 digits; I_Φ is the finite sum (3.7). The total numerical uncertainty in C_2 is below 10^{-9} .

The margins $C_0 - C_2 \in \{2.14, 4.55, 3.80\}$ exceed the total numerical uncertainty by more than nine orders of magnitude; the inequalities (P2)–(P4) hold with slack at least 0.18 (dominant mode) and 28 (subdominant modes) on the finite parts and by thousands on the tails. The verification takes about one minute on a laptop.

7. FURTHER INSTANCES: $r = 9, 11, 13, 15, 21$

The same programme (Sections 3–5, run by the same code with the same certificates (P1)–(P5)) proves the following statements; nothing in the proofs of Sections 2–4 depends on $r \in \{3, 5, 7\}$, and the certificates below were produced exactly as in Section 5. The only difference is technical: for $q > 60$ the saddle points of all modes are located by Newton’s method with continuation in the mode parameter (starting from the dominant one) instead of by the roots of the polynomial P , since the certificates only use the fact that u_k are critical points of F_k (residuals $< 10^{-40}$), and the rays are followed to $|\operatorname{Im} u| = Y$ with Y doubled until the closed-form tail bounds of Lemma 4.9 fall below the threshold (they lose $O(q\eta_0)$ against the truth).

Theorem 7.1 ($r = 9$). *At least one of the eighteen numbers $\zeta(11), \zeta(13), \dots, \zeta(45)$ is irrational.*

Theorem 7.2 ($r = 11$). *At least one of the twenty-two numbers $\zeta(13), \zeta(15), \dots, \zeta(55)$ is irrational.*

Theorem 7.3 ($r = 13$). *At least one of the twenty-six numbers $\zeta(15), \zeta(17), \dots, \zeta(65)$ is irrational.*

Theorem 7.4 ($r = 15$). *At least one of the thirty numbers $\zeta(17), \zeta(19), \dots, \zeta(75)$ is irrational.*

Theorem 7.5 ($r = 21$). *At least one of the forty-two numbers $\zeta(23), \zeta(25), \dots, \zeta(105)$ is irrational.*

For every odd $r \leq 25$ these (with Theorems 1.1–1.3 and the numerically established but not yet certified cases $r = 17, 19, 23, 25$ of Table 4) give “one of $\zeta(r+2), \dots, \zeta(5r)$ ”; for comparison, Zudilin’s uniform Theorem 0.2 in [12] gives “one of $\zeta(b+2), \dots, \zeta(8b-1)$ ” for every odd b , i.e. $\zeta(11), \dots, \zeta(71)$ for $b = 9$ and $\zeta(23), \dots, \zeta(167)$ for $b = 21$.

The direction vectors (exponents denote multiplicities; the semicolon separates zero blocks from pole blocks) are

- (D) $r = 9$, $q = 47$, $\eta_0 = 150$, $\boldsymbol{\eta}' = (44^9; 46^2, 47^2, 48^2, 49^3, 50^2, 51^2, 52^3, 53^2, 54^2, 55^2, 56^3, 57^2, 58^2, 59^3, 60^2, 61^2, 62^2)$, $\sum \eta_j = 2448 < 2850$;
- (E) $r = 11$, $q = 57$, $\eta_0 = 150$, $\boldsymbol{\eta}' = (44^{11}; 46^2, 47^3, 48^3, 49^2, 50^3, 51^3, 52^3, 53^3, 54^2, 55^3, 56^3, 57^3, 58^3, 59^2, 60^3, 61^3, 62^2)$, $\sum \eta_j = 2968 < 3450$;
- (F) $r = 13$, $q = 67$, $\eta_0 = 150$, $\boldsymbol{\eta}' = (46^{13}; 48^9, 49^5, 50^6, 51^4, 52^3, 53^4, 54^3, 55^4, 56^3, 57^3, 58^4, 59^3, 60^3)$, $\sum \eta_j = 3457 < 4050$;
- (G) $r = 15$, $q = 77$, $\eta_0 = 150$, $\boldsymbol{\eta}' = (44^{15}; 47^{10}, 48^3, 49^4, 50^4, 51^3, 52^4, 53^3, 54^3, 55^4, 56^3, 57^3, 58^4, 59^3, 60^3, 61^4, 62^3, 63)$, $\sum \eta_j = 3993 < 4650$;
- (H) $r = 21$, $q = 107$, $\eta_0 = 150$, $\boldsymbol{\eta}' = (44^{21}; 47^{14}, 48^5, 49^5, 50^6, 51^5, 52^4, 53^5, 54^5, 55^4, 56^5, 57^5, 58^4, 59^5, 60^5, 61^4, 62^5)$, $\sum \eta_j = 5531 < 6450$;

(D) and (E) are “uniform” families (pole directions increasing by $16/(q-r-1)$ per block from 46 to 62, rounded); (F), (G), (H) were found by a hill climb over the multiplicities of the pole directions, which favours a cluster of poles just above the zero direction.

TABLE 3. Certified data for Theorems 7.1–7.5. Notation as in Tables 1 and 2; $\theta = H_d - \delta/2$ is the threshold used by the program (a slightly stronger requirement than $H_d - \delta$ in (P3)–(P4)). “sup” entries are certified suprema (Lemma 4.8) over the parts of $\mathcal{L}$ with $|\operatorname{Im} u| \leq Y$, “tail” entries are $\beta_{\pm}(Y)$; all subdominant modes $k < k_d$ are listed by their worst certified value on $\mathcal{L}$ (all tails are far below θ).

	(D)	(E)	(F)	(G)	(H)
r, q, k_d	9, 47, 7	11, 57, 9	13, 67, 11	15, 77, 13	21, 107, 19
$u_* = u_{k_d}$	3.797806 – 5.563235i	4.926286 – 5.327998i	5.422706 – 4.569219i	6.198865 – 4.482888i	7.128273 – 3.517974i
$C_0 = -H_{k_d}(u_*)$	1600.2358611575	1919.0089806461	2300.4945489120	2566.5247868004	3553.0267564243
$\alpha_{k_d} \bmod \pi$	0.771701	2.902079	0.353392	2.800534	0.559820
$\phi; f(u_*) $	2.50333; 1.08799	2.58389; 1.16238	2.66481; 1.36377	2.70514; 1.40076	2.81491; 1.79110
$M_3; 3 f /(2M_3)$	0.2946; 5.54	0.3056; 5.70	0.3789; 5.40	0.3714; 5.66	0.4785; 5.61
$s_0; s_1; x_0; Y$	1; 1; –2; 1200	1; 1; –2; 1200	1; 1; –2; 1200	1; 2; –2; 2400	1; 3; –2; 2400
$\theta = H_d - \delta/2$	–1600.371859	–1919.154279	–2300.665020	–2566.699882	–3553.250644
top	2.9947 – 4.9674i	4.0778 – 4.7988i	4.5342 – 4.1103i	5.2926 – 4.0602i	6.1812 – 3.1971i
ext	5.4041 – 6.7548i	6.6232 – 6.3865i	7.1997 – 5.4871i	8.9176 – 5.7511i	10.9167 – 4.8016i
sup on [top, x_0]	–1600.807960	–1919.616419	–2301.208714	–2567.252674	–3553.952684
sup on upper ray	–1672.175905	–2016.227099	–2416.899947	–2710.069266	–3764.148412
sup on [ext, bot]	–1600.747634	–1919.556363	–2301.135192	–2567.182148	–3553.863701
sup on lower ray	–1602.136837	–1921.038668	–2302.863635	–2571.749423	–3559.632787
tails $\beta_+(Y), \beta_-(Y)$	–59052, –5093	–73813, –4219	–88775, –3624	–210084, –10800	–300176, –8156
$\max_{k < k_d} \sup_{\mathcal{L}} H_k$	–1628.506868	–1947.538326	–2326.069365	–2592.109408	–3574.017575
$\min_k \operatorname{dist}(\alpha_k, \pi\mathbb{Z})$	0.621088	0.239514	0.062068	0.027188	0.164194
$m_1; \min_j m_j; D\text{-part}$	60; 58; 2690	60; 58; 3275	56; 54; 3611	59; 56; 4338	59; 56; 6053
$\#\text{breakpoints}; \max \varphi$	2323; 32	2323; 39	1951; 50	2367; 55	2275; 76
I_{Φ}	1130.2413903609	1372.1567283459	1367.4109875263	1816.7465626742	2511.9175820351
C_2	1559.7586096391	1902.8432716541	2243.5890124737	2521.2534373258	3541.0824179649
$C_0 - C_2$	+40.477252	+16.165709	+56.905536	+45.271349	+11.944338

7.1. The statement “one of $\zeta(r+2), \dots, \zeta(5r)$ ” and the limit $r \rightarrow \infty$. Since the length of the lists grows linearly in r (Theorems 1.1–7.5: 21, 33, 45, 55, 65, 75, 105 for $r = 5, \dots, 21$), it is natural to ask whether “at least one of $\zeta(r+2), \zeta(r+4), \dots, \zeta(5r)$ is irrational” holds for every odd r , i.e. whether $q = 5r + 2$ is always admissible. Table 4 records what the construction gives for $q = 5r + 2$ after optimising the direction vector (multiplicities of integer pole directions at $\eta_0 = 150$, zero blocks all equal), together with the smallest q for which a positive margin was found.

Thus the “5r” statement is proved for all odd $r \leq 25$ (certified for $r \leq 15$ and $r = 21$, numerically for $r = 17, 19, 23, 25$), and it is *not* provable by this family of linear forms for $r = 31$ or $r = 61$ at resolution $\eta_0 = 150$. The deficits grow linearly in r , and the reason is visible in the limit $r \rightarrow \infty$, which is simpler than the finite case.

The limit $r \rightarrow \infty$. Fix η_0 , one zero direction z and pole directions with multiplicities proportional to r : $P = q - r = \lambda r$ poles, a fraction w_p of them at the integer p ($\sum_p w_p = \lambda$). Then $f = r\bar{f}$ with

$$\begin{aligned} \bar{f}(u) = & (-u) \log(-u) + (\eta_0 + u) \log(\eta_0 + u) + [(z + u) \log(z + u) - (\eta_0 - z + u) \log(\eta_0 - z + u) - 2z \log z] \\ & + \sum_p w_p [(p + u) \log(p + u) - (\eta_0 - p + u) \log(\eta_0 - p + u) + (\eta_0 - 2p) \log(\eta_0 - 2p)], \end{aligned}$$

and the mode equation $f'(u) = -i(r-2)\pi$ reads $\bar{f}'(u) = -i\pi(1 - 2/r)$. On the positive real axis approached from below, $\operatorname{Im} \bar{f}'(x - i0) = -\pi$ exactly (from the term $-\log(-u)$), so the dominant saddle collapses onto the real axis: if $a > 0$ is the root of $\operatorname{Re} \bar{f}'(a) = 0$ (it exists because $\operatorname{Re} \bar{f}' \rightarrow +\infty$ at 0^+ and $\operatorname{Re} \bar{f}'(x) \sim 2[z - \lambda(\frac{1}{2} - \sum_p w_p p)]/x < 0$ as $x \rightarrow \infty$, by the degree condition (1.1))

TABLE 4. The case $q = 5r + 2$ (list $\zeta(r+2), \dots, \zeta(5r)$). “margin” is $C_0 - C_2$ for the best direction vector found ($\eta_0 = 150$ unless stated; by the exact scaling law, margins at different η_0 are compared per unit of η_0). Rows marked $\checkmark$ are certified (Theorems 1.1–7.5); the others are numerical values from the same code, not yet run through the certificate program. The last column is the best q found overall, with $(q-2)/r$.

r	$q = 5r + 2$	C_0	margin at $q = 5r + 2$		best q found	$((q-2)/r)$
3	17	—	implied by $\zeta(5), \dots, \zeta(11)$	$\checkmark$	13	(3.67)
5	27	—	implied by Theorem 1.1	$\checkmark$	23	(4.20)
7	37	—	implied by Theorem 1.2	$\checkmark$	35	(4.71)
9	47	1600.2	+40.48	$\checkmark$	47	(5.00)
11	57	1919.0	+16.17	$\checkmark$	57	(5.00)
13	67	2300.5	+56.91	$\checkmark$	67	(5.00)
15	77	2566.5	+45.27	$\checkmark$	77	(5.00)
17	87	2851	+35.13		87	(5.00)
19	97	3123	+31.73		97	(5.00)
21	107	3556	+11.94	$\checkmark$	107	(5.00)
23	117	3812	+5.61		117	(5.00)
25	127	4180	−3.03; +25.20 at $\eta_0 = 300$		127	(5.00)
31	157	5233	−50.75		161	(5.13)
61	307	10266	−240.42		≤ 325	(≤ 5.30)

and $D = -f''(a) = r\bar{D}(a) > 0$, then expanding $F_{r-2}(u) = f(u) + i(r-2)\pi u$ around a (where $f'(a-i0) = -ir\pi$) gives $\operatorname{Re} F_{r-2}(a-iy) = \operatorname{Re} f(a) - 2\pi y + \frac{1}{2}Dy^2 + O(ry^3)$, hence

$$(7.1) \quad u_{r-2} = a - \frac{2\pi i}{D} + O(r^{-2}), \quad C_0 = -\operatorname{Re} f(a) + \frac{2\pi^2}{D} + O(r^{-2}), \quad \frac{2\pi^2}{D} = \frac{2\pi^2}{r\bar{D}(a)} = O(r^{-1}).$$

(Numerically, for the optimised vectors: the correction $2\pi^2/D$ is 27.7, 15.7, 10.6, 6.5, 4.8, 3.3 for $r = 9, 13, 21, 31, 41, 61$, and (7.1) reproduces the Newton saddle value of C_0 to 6.0, 1.7, 0.5, 0.14, 0.06, 0.02.) So the “derivative penalty” disappears as $r \rightarrow \infty$; what remains is a competition that is linear in r . Indeed Δ_n has $q-1$ factors $D_{\hat{m}n}$, so

$$(7.2) \quad C_0 - C_2 = r\delta(z, w; \lambda) + \hat{m}_P + O(r^{-1}), \quad \delta = -\operatorname{Re} \bar{f}(a) - \left(\hat{m}_1 + \sum_p w_p \hat{m}(p) - \bar{I}_\Phi(z, w) \right),$$

where $\hat{m}_1 = \max(z, \eta_0 - 2p_{\min}, \eta_0 - z - p_{\min})$, $\hat{m}(p) = \max(z, \eta_0 - 2p_{\min}, \eta_0 - z - p)$, $\bar{I}_\Phi$ is the integral of Section 3 computed with $\varphi/r = \min_y \sum(\text{rows}) \cdot [\cdot]$ with the real weights 1 (zero rows) and w_p (pole rows), and $\hat{m}_P \approx 0.37\eta_0$ is the single saved factor. Everything in δ is independent of r and computable by the code of Section 3 with real weights. Maximising δ over z and w (weights on the integers $z \leq p < \eta_0/2$, $\eta_0 = 150$, hill climb with mass transfers) gives

$$\max \delta(\cdot; \lambda) \approx -6.00 \ (\lambda = 4), \quad -0.61 \ (\lambda = 4.15), \quad +4.04 \ (\lambda = 4.30),$$

attained at $z = 46$ with a cluster of weight ≈ 0.68 at $p = 48$ and an almost uniform density ≈ 0.24 on $49 \leq p \leq 60$; allowing two zero directions changes $\delta(\cdot; 4)$ by less than 1, and doubling the resolution ($\eta_0 = 300$) by about +1 per unit r (per $\eta_0 = 150$). Hence

$$(7.3) \quad \lambda_* := \inf \{ \lambda : \max_{z, w} \delta(z, w; \lambda) > 0 \} \approx 4.15 \ (\pm 0.05), \quad \text{i.e.} \quad \frac{q_{\min}(r)}{r} \longrightarrow 1 + \lambda_* \approx 5.15.$$

Consequently the construction proves “one of $\zeta(r+2), \dots, \zeta(q-2)$ ” for all large r exactly when $q \geq (1 + \lambda_*)r + O(1)$; the statement with $5r$ ($\lambda = 4 + 2/r$) has margin $\approx r\delta(4) + 2\delta'(4) + \hat{m}_P \approx 122 - 6.0r$ at $\eta_0 = 150$, in agreement with Table 4, and fails for every $r \gtrsim 27$. The correct asymptotic form of

the family of theorems is therefore “one of $\zeta(r+2), \dots, \zeta(\lceil(1+\lambda_*)r\rceil + O(1))$ ” with $1+\lambda_* \approx 5.15$, to be compared with 8 in Zudilin’s Theorem 0.2 [12]. A uniform-in- r theorem is proved in Section 8: since the criterion is applied for each fixed r as $n \rightarrow \infty$, no uniformity of Lemmas 4.3 and 4.5 in r is needed; what has to be uniform is only the verification of (P1)–(P5) and of $C_0 > C_2$, and for a family with integer weights w_p this reduces to finitely many inequalities about $\tilde{f}$ that do not involve r . With integer weights the smallest admissible λ is 5 (i.e. $q = 6r + 1$), which gives the constant 6 there; the constant $1 + \lambda_* \approx 5.15$ would need rational weights and a rounding scheme, see Remark 8.9.

8. A UNIFORM THEOREM: ONE OF $\zeta(r+2), \dots, \zeta(6r-1)$ FOR EVERY ODD r

Theorem 8.1. *For every odd integer $r \geq 3$, at least one of the $\frac{5r-1}{2}$ numbers $\zeta(r+2), \zeta(r+4), \dots, \zeta(6r-1)$ is irrational.*

This sharpens Zudilin’s Theorem 0.2 of [12] (“one of $\zeta(b+2), \dots, \zeta(8b-1)$ ” for every odd b) from $8r-1$ to $6r-1$. The proof has three layers. For $r \leq 15$ the lists of Theorems 1.3, 1.1, 1.2, 7.1–7.4 are contained in $\{\zeta(r+2), \dots, \zeta(6r-1)\}$. For $17 \leq r \leq 49$ we certify, exactly as in Sections 5 and 7, one direction vector for each r (Table 6); the vectors are the members $\boldsymbol{\eta}^{(r)}$ of the family (8.1) below, and the certification is cheap because the family has only seven distinct directions. For $r \geq 51$ the family (8.1) is treated *uniformly*: the phase function is r times a fixed function g plus an r -independent perturbation, the dominant saddle point collapses onto the real axis at rate $1/r$ (as explained in §7.1), and all inequalities (P1)–(P5) reduce to finitely many inequalities about g which do not depend on r , plus scalar inequalities in r that are checked for $51 \leq r \leq 19999$ one by one and for $r > 19999$ by explicit asymptotic bounds.

8.1. The family and the structure of the phase function. Let $\eta_0 = 150$, $z = 45$, $\mathcal{P} = \{48, 50, 53, 56, 60\}$, $p_e = 74$, and for odd $r \geq 3$ let

$$(8.1) \quad \boldsymbol{\eta}^{(r)} = (150; 45^r; 48^r, 50^r, 53^r, 56^r, 60^r, 74), \quad q = q_r = 6r + 1,$$

i.e. r zero blocks with direction 45, r pole blocks with each of the directions 48, 50, 53, 56, 60, and one pole block with direction 74. Condition (1.1) holds: $45 \leq 48 \leq \dots \leq 74 < 75$ and $\sum_j \eta_j = 312r + 74 < 75(5r + 1) = \frac{1}{2}\eta_0(q - r)$. The linear form $F_n(\boldsymbol{\eta}^{(r)})$ lies in $\mathbb{Q} + \mathbb{Q}\zeta(r+2) + \dots + \mathbb{Q}\zeta(6r-1)$ (Lemma 2.2).

For $u \in \mathbb{C} \setminus (-\infty, 0]$, with principal logarithms, put

$$\begin{aligned} Z(u) &= (z+u) \log(z+u) - (\eta_0 - z + u) \log(\eta_0 - z + u) - 2z \log z, \\ \Pi_p(u) &= (p+u) \log(p+u) - (\eta_0 - p + u) \log(\eta_0 - p + u) + (\eta_0 - 2p) \log(\eta_0 - 2p), \\ g(u) &= -u \log u + (\eta_0 + u) \log(\eta_0 + u) + Z(u) + \sum_{p \in \mathcal{P}} \Pi_p(u), \quad e(u) = \Pi_{74}(u), \end{aligned}$$

$$F(u) = r g(u) + e(u) - 2\pi i u.$$

g and e are analytic on $\mathbb{C} \setminus (-\infty, 0]$ and real on $(0, \infty)$; for real $x > -z$ let $\hat{g}(x) = -x \log |x| + (\eta_0 + x) \log(\eta_0 + x) + Z(x) + \sum_p \Pi_p(x)$ (continuous on $(-z, \infty)$, equal to g on $(0, \infty)$ and to $\operatorname{Re} g(x \pm i0)$ on $(-z, 0)$). Since $\Pi'_p(x) = \log \frac{p+x}{\eta_0-p+x} < 0$, the functions Π_p and e are decreasing on $(-p, \infty)$. We write $g'(u) = \sum_b c_b \log(b+u)$ with the coefficient list

$$(8.2) \quad \begin{aligned} (b, c_b) &\in \{(0, -1), (\eta_0, 1), (z, 1), (\eta_0 - z, -1)\} \cup \{(p, 1), (\eta_0 - p, -1) : p \in \mathcal{P}\}, \\ \sum_b c_b &= 0, \quad \sum_b c_b b = c_g := -126, \end{aligned}$$

so that $g^{(k)}(u) = (-1)^k (k-2)! \sum_b c_b (b+u)^{1-k}$ for $k \geq 2$, and similarly $e'(u) = \log(74+u) - \log(76+u)$.

Lemma 8.2 (structure). *Let f , $F_k = f + ik\pi u$, $H_k = \operatorname{Re} F_k$ be the functions of Section 4 for $\eta = \eta^{(r)}$. Then*

$$F_k(u) = r g(u) + e(u) - i(r-k)\pi u \quad (\operatorname{Im} u < 0), \quad F_k(u) = r g(u) + e(u) + i(r+k)\pi u \quad (\operatorname{Im} u > 0).$$

Consequently $F_{r-2} = F$ on the lower half plane, and for every $u \in \Omega$:

- (a) if $\operatorname{Im} u < 0$ and $k \leq r-2$: $H_k(u) = r \operatorname{Re} g(u) + \operatorname{Re} e(u) - (r-k)\pi |\operatorname{Im} u| \leq H_{r-2}(u) - (r-2-k)\pi |\operatorname{Im} u|$;
- (b) if $\operatorname{Im} u > 0$ and $k \geq 1$: $H_k(u) = r \operatorname{Re} g(u) + \operatorname{Re} e(u) - (r+k)\pi \operatorname{Im} u \leq r \operatorname{Re} g(u) + \operatorname{Re} e(u) - (r+1)\pi \operatorname{Im} u$;
- (c) if $u = x \in (-z, 0)$: $H_k(x) = r\hat{g}(x) + e(x)$ for every k .

Proof. For $\operatorname{Im} u < 0$ the principal branches satisfy $\log(-u) = \log u + i\pi$, hence $(-u) \log(-u) = -u \log u - i\pi u$; for $\operatorname{Im} u > 0$, $\log(-u) = \log u - i\pi$. Inserting this in (4.1) and collecting the blocks (κ_0 is the sum of the constants in Z and the Π_p) gives $f(u) = r(g(u) \mp i\pi u) + e(u)$ for $\pm \operatorname{Im} u > 0$, and the displayed formulas follow. (a), (b) are immediate ($\operatorname{Re}(-i(r-k)\pi u) = (r-k)\pi \operatorname{Im} u = -(r-k)\pi |\operatorname{Im} u|$ for $\operatorname{Im} u < 0$), and (c) follows from either formula by continuity, since $\operatorname{Re}(\pm i\pi ru) = 0$ for real u . $\square$

Lemma 8.3 (reduction of (P4)). *In Proposition 4.7, hypothesis (P4) may be replaced by*

$$(P4') \quad \sigma \subset \{\operatorname{Im} u \leq -y_\sigma\} \text{ for some } y_\sigma > 0, \text{ and } H_1(u) \leq H_d - \delta \text{ for all } u \in \mathcal{L} \text{ with } \operatorname{Im} u \geq 0;$$

the conclusion (4.10) then holds with the error term $O(n\mathcal{A}_n e^{n(H_d - \delta'')})$, $\delta'' = \min(\delta, 2\pi y_\sigma) > 0$, in its proof, hence unchanged.

Proof. Let $k \in \{1, 3, \dots, r-4\}$. On $\mathcal{L} \cap \{\operatorname{Im} u < 0\} \setminus \sigma$, Lemma 8.2(a) and (P3) give $H_k \leq H_{r-2} \leq H_d - \delta$; on σ , Lemma 8.2(a) and (P2) give $H_k \leq H_{r-2} - 2\pi |\operatorname{Im} u| \leq H_d - 2\pi y_\sigma$; on $\mathcal{L} \cap \{\operatorname{Im} u \geq 0\}$, Lemma 8.2(b),(c) give $H_k \leq H_1 \leq H_d - \delta$. So (P4) holds with δ'' in place of δ , which is all that the proof of Proposition 4.7 uses. $\square$

8.2. The arithmetic side.

Lemma 8.4. *For $\eta^{(r)}$: $\hat{m}_0 = 54$, $\hat{m}_1 = 57$, the $\hat{m}_j$ ($j \geq 2$) take the values 57 ($r-1$ times), 55 (r times) and 54 ($3r+1$ times), and with $C_2 = C_2(r)$ of Lemma 3.3,*

$$C_2(r) \leq \bar{c}_2 r - 3, \quad \bar{c}_2 = 331 - \bar{I}_\Phi = 214.6595862475, \quad \bar{I}_\Phi = 116.3404137525.$$

Here $\bar{I}_\Phi$ is the integral (3.6) for the direction vector (150; 45; 48, 50, 53, 56, 60) with $r = 1$ (that is, φ replaced by $\bar{\varphi} = \min_y \bar{\varphi}_0$, where $\bar{\varphi}_0$ is (3.2) for that vector), and $\Delta_n \in \mathbb{Z}$.

Proof. From (3.1): $\hat{m}_0 = \max(45, 150 - 96) = 54$ and $\hat{m}_j = \max(54, 105 - p_{(j)})$ where $p_{(1)} \leq p_{(2)} \leq \dots$ are the pole directions of $\eta^{(r)}$; this gives the stated values and $r\hat{m}_1 + \sum_{j \geq 2} \hat{m}_j = 57r + 57(r-1) + 55r + 54(3r+1) = 331r - 3$. Next, φ_0 of (3.2) for $\eta^{(r)}$ equals $r\bar{\varphi}_0(x, y) + \rho(x, y)$, where $\rho \in \{0, 1\}$ is the summand of the block with direction 74 (Lemma 3.1); hence $\varphi(x) = \min_y \varphi_0(x, y) \geq r\bar{\varphi}(x)$, and since $\psi'(1+x) > 0$ and $\hat{m}_{q-r} = 54$ for both vectors, $I_\Phi \geq r\bar{I}_\Phi$. Thus $C_2(r) = 331r - 3 - I_\Phi \leq (331 - \bar{I}_\Phi)r - 3$. Finally $\varphi \leq r \max \bar{\varphi} + 1 = 4r + 1 < q - 1$, so $\Delta_n \in \mathbb{Z}$ by the last sentence of the proof of Lemma 3.2. The value of $\bar{I}_\Phi$ was computed as in §3.3 (1211 breakpoints, $\max \bar{\varphi} = 4$, exact-rational recheck of $\bar{\varphi}$ on every interval, ψ to 30 digits). $\square$

8.3. Constants. The function $\hat{g}'(x) = \log \frac{\eta_0 + x}{x} + \log \frac{z+x}{\eta_0 - z+x} + \sum_p \log \frac{p+x}{\eta_0 - p+x}$ tends to $+\infty$ as $x \rightarrow 0^+$ and behaves like $c_g/x < 0$ as $x \rightarrow \infty$. It changes sign in the interval $[a_-, a_+] = [a_0 - 10^{-30}, a_0 + 10^{-30}]$, $a_0 = 4.19295271437465633113106378004886$ (both signs verified at 40 digits), and $\hat{g}'' < -0.18$ there, so there is a unique zero a of $\hat{g}'$ in $[a_-, a_+]$; all quantities below that depend

on a are evaluated at the midpoint, with an error below 10^{-28} which is irrelevant for every inequality that follows (each holds with slack at least 10^{-3}). We use

$$\begin{aligned}
 (8.3) \quad & a = 4.192952714375, \quad \hat{g}(a) = -232.0454847547, \quad \bar{D} := -\hat{g}''(a) = 0.1832488610, \\
 & g_3 := \hat{g}'''(a) = 0.0554715328, \quad e(a) = -9.3574271140, \\
 & e_1 := e'(a) = -0.0252561149, \quad e_2 := e''(a) = 0.0003189526, \\
 & \nu := \frac{\sqrt{4\pi^2 + e_1^2}}{\bar{D}} = 34.28799521, \quad \kappa_2 := \frac{g_3(e_1 - 2\pi i)}{\bar{D}} = -0.00764532 - 1.90199228i, \\
 & \hat{g}(x_0) = -238.59979763, \quad e(x_0) = -9.19456197 \quad (x_0 = -2),
 \end{aligned}$$

and the following bounds, valid because $|b + u| \geq b + \operatorname{Re} u$ whenever $b + \operatorname{Re} u > 0$: for $\tau \in (0, a)$ and $k \geq 2$,

$$\begin{aligned}
 M_k(\tau) &:= (k-2)! \sum_b \frac{|c_b|}{(b+a-\tau)^{k-1}} \geq \sup_{|u-a| \leq \tau} |g^{(k)}(u)|, \\
 M_k^{\text{hp}}(\xi) &:= (k-2)! \sum_b \frac{|c_b|}{(b+\xi)^{k-1}} \geq \sup_{\operatorname{Re} u \geq \xi} |g^{(k)}(u)| \quad (\xi > 0),
 \end{aligned}$$

$E_k(\tau)$, $E_k^{\text{hp}}(\xi)$ the same sums over $\{(74, 1), (76, -1)\}$ for e , and $E_1(\tau) := |e_1| + E_2(\tau)\tau$, which bounds $|e'|$ on $|u - a| \leq \tau$. Since $g'(a) = 0$, also $|g(u) - g(a)| \leq \frac{1}{2}M_2(\tau)\tau^2$ for $|u - a| \leq \tau$.

Lemma 8.5 (saddle point). *Let $\tilde{u} \in \mathbb{C}$ and $\rho > 0$ with $\xi := \operatorname{Re} \tilde{u} - \rho > 0$, and put $\beta = 1/|F''(\tilde{u})|$, $\eta = \beta|F'(\tilde{u})|$, $L = rM_3^{\text{hp}}(\xi) + E_3^{\text{hp}}(\xi)$. If $h := \beta L\eta \leq \frac{1}{2}$ and $2\eta \leq \rho$, then F' has a zero u_* with $|u_* - \tilde{u}| \leq 2\eta$.*

Proof. F' is analytic on $\{\operatorname{Re} u > 0\} \supset \bar{B}(\tilde{u}, \rho)$, F'' has Lipschitz constant $\sup |F'''| \leq L$ there, and $F''(\tilde{u}) \neq 0$. This is the Newton–Kantorovich theorem ([7, Thm. 12.6.2], applied in the Banach space $\mathbb{C}$): the Newton sequence from $\tilde{u}$ converges to a zero u_* in $\bar{B}(\tilde{u}, t_*)$, $t_* = (1 - \sqrt{1 - 2h})/(\beta L) \leq 2\eta \leq \rho$. $\square$

8.4. The path. Fix $s_0 = 1.5$, $x_0 = -2$, $y_c = 0.25$, $x_L = a - s_0 - 0.6 = 2.092953$, $x_R = 20$, $P_L = x_L - iy_c$, $P_R = x_R - iy_c$, and the rectangles

$$B_L = [2.092953, 2.702953] \times [-0.523633, 0], \quad B_R = [5.604839, 20] \times [-0.827198, 0].$$

For $r \geq 51$ let u_* be the zero of F' provided by Lemma 8.5 (§8.6, §8.7), ϕ the solution of $e^{2i\phi}F''(u_*) = -|F''(u_*)|$ with $\operatorname{Re} e^{i\phi} < 0$, $e = e^{i\phi}$, $\text{top} = u_* + s_0e$, $\text{bot} = u_* - s_0e$, $\sigma = [\text{bot}, \text{top}]$, and

$$(8.4) \quad \mathcal{L}_r : -i\infty \xrightarrow{\operatorname{Re} u = x_R} P_R \longrightarrow \text{bot} \xrightarrow{\sigma} \text{top} \longrightarrow P_L \longrightarrow x_0 \xrightarrow{\operatorname{Re} u = x_0} +i\infty.$$

We shall show $\text{top} \in B_L$, $\text{bot} \in B_R$ with negative imaginary parts; then $[P_R, \text{bot}] \subset B_R$, $[\text{top}, P_L] \subset B_L$ (convexity), all of $\mathcal{L}_r$ except x_0 lies in $\{\operatorname{Im} u \neq 0\}$, $\inf_{\mathcal{L}_r} \operatorname{Re} u = x_0 > -z$, $\inf_{\mathcal{L}_r} |u| \geq \min(2, y_c|x_0|/|P_L - x_0|) > 0.12$ (the distance from 0 to the segment $[P_L, x_0]$ is $y_c|x_0|/|P_L - x_0| = 0.1219$; all other pieces have $\operatorname{Re} u \geq x_L$ or $|u| \geq |x_0|$), and $\sup_{\mathcal{L}_r} |\arg(-u)| < \pi$ (the path is a finite union of segments and two rays on which $|\arg(-u)| < \pi$ is continuous and tends to $\pi/2$ at infinity). So (P5) and the hypotheses of Lemma 4.6 hold for every such r .

Lemma 8.6 (monotonicity on the rays). (i) *For $y > 0$: $\frac{d}{dy}[\operatorname{Re} g(x_0 + iy) - \pi y] < 0$ and $\frac{d}{dy}[\operatorname{Re} e(x_0 + iy) - \pi y] < 0$.* (ii) *For $x > 0$ and $y > 0$: $\frac{d}{dy} \operatorname{Re} e(x - iy) < 0$, and $\frac{d}{dy} \operatorname{Re} g(x - iy) \leq \frac{c_g}{y} + \frac{S_3(x)}{3y^3}$ with $S_3(x) = \sum_{c_b < 0} (b+x)^3$; in particular $y \mapsto \operatorname{Re} g(x - iy)$ is decreasing on $[Y_1(x), \infty)$, $Y_1(x) = \sqrt{S_3(x)/(3|c_g|)}$. For $x = x_R$: $S_3 = 9919130$, $Y_1 = 161.991$.*

Proof. $\frac{d}{dy} \operatorname{Re} g(x + iy) = -\operatorname{Im} g'(x + iy)$ and $\frac{d}{dy} \operatorname{Re} g(x - iy) = \operatorname{Im} g'(x - iy)$, with $\operatorname{Im} g'(u) = \sum_b c_b \arg(b + u)$. (i) For $u = x_0 + iy$, $x_0 < 0 < y$: $\arg u = \pi - \arctan(y/|x_0|)$ and $\arg(b + u) = \arctan \frac{y}{b+x_0} \in (0, \frac{\pi}{2})$ for $b > 0$. Hence $\frac{d}{dy} [\operatorname{Re} g - \pi y] = -\arctan \frac{y}{|x_0|} - A(y)$ with $A(y) = \arctan \frac{y}{\eta_0+x_0} + [\arctan \frac{y}{z+x_0} - \arctan \frac{y}{\eta_0-z+x_0}] + \sum_p [\arctan \frac{y}{p+x_0} - \arctan \frac{y}{\eta_0-p+x_0}] \geq 0$, because $z < \eta_0 - z$ and $p < \eta_0 - p$. For e : $\frac{d}{dy} [\operatorname{Re} e - \pi y] = -[\arctan \frac{y}{74+x_0} - \arctan \frac{y}{76+x_0}] - \pi < 0$. (ii) For $u = x - iy$ and $b + x > 0$: $\arg(b + u) = -\arctan \frac{y}{b+x} = -\frac{\pi}{2} + \arctan \frac{b+x}{y}$; as $\sum c_b = 0$, $\operatorname{Im} g'(u) = \sum_b c_b \arctan \frac{b+x}{y}$. Using $s - s^3/3 \leq \arctan s \leq s$ ($s \geq 0$) for the terms with $c_b > 0$ and $c_b < 0$ respectively, $\operatorname{Im} g'(u) \leq \frac{1}{y} \sum_b c_b (b+x) + \frac{1}{3y^3} \sum_{c_b < 0} (b+x)^3 = \frac{c_g}{y} + \frac{S_3}{3y^3}$, which is negative for $y > Y_1$. Finally $\frac{d}{dy} \operatorname{Re} e(x - iy) = \arctan \frac{74+x}{y} - \arctan \frac{76+x}{y} < 0$. $\square$

Lemma 8.7 (*r-independent bounds*). *The following suprema hold (certified as in Lemma 4.8: grid values plus Lipschitz corrections; for the rectangles the grid has mesh $2 \cdot 10^{-4}$ on B_L and on $B_R \cap \{\operatorname{Re} u \leq 7.605\}$, 10^{-3} on the rest of B_R , and the Lipschitz constant of $\operatorname{Re} g$, resp. of $\operatorname{Re} e + 2\pi \operatorname{Im} u$, is bounded by $\max_{\text{grid}} |g'| + \frac{h}{\sqrt{2}} \sup |g''|$, resp. $\max_{\text{grid}} |e'| + \frac{h}{\sqrt{2}} \sup |e''| + 2\pi$, with $\sup |g''|, \sup |e''|$ from the distances of the rectangle to the branch points; segments as in Lemma 4.8 with mesh 10^{-4} , resp. $2 \cdot 10^{-3}$ on the lower ray):*

piece	$S_g := \sup \operatorname{Re} g$	$S_e := \sup (\operatorname{Re} e + 2\pi \operatorname{Im} u)$	$\Gamma := \hat{g}(a) - \bar{D}s_0^2/8 - S_g$
B_L	-232.243990	-9.302985	0.146966
B_R	-232.163207	-9.391867	0.066184
$[P_L, x_0]$	-232.553934	-9.194246	0.456911
$\{x_R - iy : y_c \leq y \leq Y_1(x_R)\}$	-240.067409	-11.285922	7.970386
$\{x_0 + iy : y \geq 0\}$ (at $y = 0$)	$\hat{g}(x_0) = -238.599798$	$e(x_0) = -9.194562$	6.502774

For the upper ray the entries are $\sup_{y \geq 0} [\operatorname{Re} g(x_0 + iy) - \pi y]$ and $\sup_{y \geq 0} [\operatorname{Re} e(x_0 + iy) - \pi y]$, attained at $y = 0$ by Lemma 8.6(i). On the lower ray, $y \mapsto r \operatorname{Re} g(x_R - iy) + \operatorname{Re} e(x_R - iy) - 2\pi y$ is decreasing for $y \geq Y_1(x_R)$ by Lemma 8.6(ii), so the supremum over the whole ray $\{y \geq y_c\}$ of H_{r-2} is bounded by $rS_g + S_e$ with the entries of the fourth row.

8.5. The scalar inequalities. Let $r \geq 51$ be odd and let u_* be a zero of F' with $\operatorname{Re} u_* - s_0 > 0$; write $\delta = |F''(u_*)|s_0^2/8$, $H_d = \operatorname{Re} F(u_*)$, $\alpha = \operatorname{Im} F(u_*)$, $y_\sigma = |\operatorname{Im} u_*| - s_0 |\sin \phi|$, $\xi_\sigma = \operatorname{Re} u_* - s_0$, $M_3^\sigma = rM_3^{\text{hp}}(\xi_\sigma) + E_3^{\text{hp}}(\xi_\sigma)$. Consider the conditions

- (S1) $\text{top} \in B_L$, $\text{bot} \in B_R$, $\operatorname{Im} \text{top} < 0$, $\operatorname{Im} \text{bot} < 0$;
- (S2) $M_3^\sigma s_0 \leq \frac{3}{2} |F''(u_*)|$;
- (S3) $y_\sigma > 0$;
- (S4) $rS_g + S_e \leq H_d - \delta$ for each of the five rows of the table in Lemma 8.7;
- (S5) $\alpha \notin \pi\mathbb{Z}$;
- (S6) $-H_d > \bar{c}_2 r - 3$.

Proposition 8.8. *If (S1)–(S6) hold for some odd $r \geq 51$ and some zero u_* of F' , then at least one of $\zeta(r+2), \dots, \zeta(6r-1)$ is irrational.*

Proof. We verify (P1)–(P5) and (P4') for $\mathcal{L} = \mathcal{L}_r$, $k_d = r - 2$, and apply Proposition 4.7 with Lemma 8.3. By Lemma 8.2, $F_{r-2} = F$ on the lower half plane, so $F'_{r-2}(u_*) = 0$ and $f''(u_*) = F''(u_*)$, which is $\neq 0$ by (S2). (S1) and §8.4 give the geometric requirements, and $\sigma \subset \{\operatorname{Im} u \leq -y_\sigma\}$ by (S3), so (P1) holds. (P2): on the disc $|u - u_*| \leq s_0$ we have $\operatorname{Re} u \geq \xi_\sigma > 0$, so $|F'''| \leq M_3^\sigma$ there and Taylor's formula gives $\operatorname{Re} F(u_* + se) \leq H_d - \frac{1}{2} |F''(u_*)|s^2 + \frac{1}{6} M_3^\sigma |s|^3 \leq H_d - \frac{1}{4} |F''(u_*)|s^2$ for $|s| \leq s_0$ by (S2). (P3): on $[P_R, \text{bot}] \subset B_R$, $[\text{top}, P_L] \subset B_L$, $[P_L, x_0]$ and the lower ray, $H_{r-2} = r \operatorname{Re} g + \operatorname{Re} e + 2\pi \operatorname{Im} u \leq rS_g + S_e \leq H_d - \delta$ by Lemmas 8.2(a), 8.7 and (S4); on the upper ray $H_{r-2} \leq H_1$ (Lemma 8.2(b)), and $H_1(x_0 + iy) \leq r[\operatorname{Re} g(x_0 + iy) - \pi y] + [\operatorname{Re} e(x_0 + iy) - \pi y] \leq$

$r\hat{g}(x_0) + e(x_0) \leq H_d - \delta$ by Lemma 8.6(i) and (S4). This is also (P4'). (P5) was checked in §8.4. Hence $\limsup \frac{1}{n} \log |F_n| \leq H_d = -C_0$ and, by (S5), $F_n \neq 0$ for infinitely many n . By Lemma 8.4 and (S6), $C_0 > C_2(r)$, and the argument of §5 concludes. $\square$

8.6. Verification for $51 \leq r \leq 19999$. For each odd r in this range the program `unif_const.py` computes an approximate zero $\tilde{u}$ of F' by Newton's method at 40 digits, starting from $a + (e_1 - 2\pi i)/(r\bar{D})$, and applies Lemma 8.5 with $\rho = 10^{-6}$: in all cases $|F'(\tilde{u})| < 9.6 \cdot 10^{-36}$, $h < 1.2 \cdot 10^{-39}$, so F' has a zero u_* with $|u_* - \tilde{u}| < 7.1 \cdot 10^{-39}$. All quantities in (S1)–(S6) are evaluated at $\tilde{u}$; since $|F'''| \leq M_3^\sigma$ near $\tilde{u}$, replacing $\tilde{u}$ by u_* changes F'' , ϕ , H_d , α , top and bot by less than 10^{-25} , which is irrelevant because every inequality holds with slack $> 10^{-3}$. The outcome for the 9975 values of r : (S1)–(S6) hold for all of them, with

$$\begin{aligned} \max_r \frac{M_3^\sigma s_0}{\frac{3}{2}|F''(u_*)|} &= 0.8012, & \min_r y_\sigma &= 0.001325 \text{ (at } r = 19999), \\ \min_r \text{dist}(\alpha, \pi\mathbb{Z}) &= 1.0530, & \min_{r, \text{rows}} [H_d - \delta - (rS_g + S_e)] &= 1.2864, \end{aligned}$$

and $-H_d - (\bar{c}_2 r - 3) \geq 17.3r$. Table 5 lists a few values; note $\text{Im } u_* \approx -2\pi/(r\bar{D}) = -34.288/r$, $|F''(u_*)| \approx r\bar{D}$, $H_d \approx r\hat{g}(a) + e(a) - 2\pi^2/(r\bar{D})$ and $\alpha \rightarrow -2\pi a \pmod{2\pi}$, in accordance with §7.1.

TABLE 5. Sample values in the per- r verification ($\theta_{\min}$ is the smallest slack in (S4) over the five rows).

r	u_*	$ F''(u_*) $	ϕ	δ	y_σ	$\text{dist}(\alpha, \pi\mathbb{Z})$	$\theta_{\min}$
51	4.122537959 – 0.665415283i	9.427686	3.040231	2.6515	0.513633	1.0530	1.2864
53	4.127608810 – 0.640773409i	9.791393	3.044029	2.7538	0.494659	1.0642	1.4980
61	4.143222827 – 0.557992693i	11.248032	3.056751	3.1635	0.430884	1.0985	2.2932
81	4.164247747 – 0.421496569i	14.897646	3.077626	4.1900	0.325612	1.1450	4.0534
101	4.174195248 – 0.338521506i	18.553313	3.090265	5.2181	0.261563	1.1673	5.6415
201	4.187866509 – 0.170446630i	36.859387	3.115781	10.3667	0.131734	1.1988	12.7944
501	4.191968844 – 0.068426305i	91.822647	3.131235	25.8251	0.052890	1.2091	32.9734
1001	4.192637459 – 0.034251245i	183.443261	3.136408	51.5934	0.026475	1.2111	66.1739
5001	4.192918040 – 0.006856111i	916.435645	3.140555	257.7475	0.005300	1.2122	330.9967
19999	4.192945378 – 0.001714468i	3664.801488	3.141333	1030.7254	0.001325	1.2123	1323.6404

8.7. The regime $r \geq 20001$. Here (S1)–(S6) follow from explicit bounds obtained by expanding around a . Put $t = 1/r \leq t_1 := 1/20001$ and

$$u_1 = a + \frac{e_1 - 2\pi i}{r\bar{D}}, \quad \tau_1 = |u_1 - a| = \nu t.$$

Since $r\bar{D}(u_1 - a) = e_1 - 2\pi i$ and $g'(u_1) = g''(a)(u_1 - a) + R_2$ with $|R_2| \leq \frac{1}{2}M_3(\tau_1)\tau_1^2$, we get $F'(u_1) = rR_2 + e'(u_1) - e'(a)$ and $F''(u_1) = rg''(u_1) + e''(u_1)$, whence

$$|F'(u_1)| \leq \frac{N_1}{r}, \quad N_1 := \frac{1}{2}M_3(\tau_1)\nu^2 + E_2(\tau_1)\nu, \quad |F''(u_1)| \geq N_2 := r\bar{D} - M_3(\tau_1)\nu - E_2(\tau_1).$$

The hypotheses of Lemma 8.5 are monotone in β and η , so we may apply it with $\tilde{u} = u_1$, the upper bounds $\beta \leq 1/N_2$, $\eta \leq \bar{\eta} := N_1/(rN_2)$, $\rho = 2\bar{\eta}$, $L = rM_3(\tau_1 + 2\bar{\eta}) + E_3(\tau_1 + 2\bar{\eta})$ (note $M_3^{\text{hp}}(\text{Re } u_1 - \rho) \leq M_3(\tau_1 + \rho)$) and $h \leq L\bar{\eta}/N_2$: it gives a zero u_* with $|u_* - u_1| \leq \varepsilon := 2\bar{\eta}$; put $\tau = \tau_1 + \varepsilon \geq |u_* - a|$. Writing $w = u_* - a$, $w_1 = u_1 - a$ and expanding g'' , g , e around a (with $g'(a) = 0$, g and e real on $(0, \infty)$, $\text{Im } w_1 = -2\pi t/\bar{D}$, $\text{Re } w_1 = e_1 t/\bar{D}$) one obtains, with $\Theta_2 := \frac{1}{2}rM_4(\tau)\tau^2 + r|g_3|\varepsilon + E_2(\tau)$:

$$(8.5) \quad F''(u_*) = -r\bar{D} + \kappa_2 + \theta_2, \quad |\theta_2| \leq \Theta_2; \quad \left| \text{Im } u_* - \frac{2\pi}{r\bar{D}} \right| \leq \varepsilon, \quad \left| \text{Re } u_* - a - \frac{e_1}{r\bar{D}} \right| \leq \varepsilon;$$

$$(8.6) \quad r\hat{g}(a) + e(a) - \Delta^- \leq H_d \leq r\hat{g}(a) + e(a) + \Delta^+,$$

$$\Delta^+ := \frac{1}{2}rM_2(\tau)\tau^2 + E_1(\tau)\tau, \quad \Delta^- := \Delta^+ + 2\pi\left(\frac{2\pi}{r\bar{D}} + \varepsilon\right);$$

$$(8.7) \quad \left| \alpha + 2\pi a + \frac{2\pi e_1}{r\bar{D}} - \frac{\alpha_2}{r^2} \right| \leq \Delta_\alpha, \quad \alpha_2 = \frac{g_3(8\pi^3 - 6\pi e_1^2)}{6\bar{D}^3},$$

$$\Delta_\alpha := \frac{1}{2}\bar{D}\varepsilon(2\nu + r\varepsilon) + \frac{1}{2}r|g_3|\varepsilon\tau^2 + \frac{1}{24}rM_4(\tau)\tau^4 + \frac{1}{2}E_2(\tau)\tau^2 + |e_1|\varepsilon + 2\pi\varepsilon.$$

(For (8.7): $\alpha = r \operatorname{Im} g(u_*) + \operatorname{Im} e(u_*) - 2\pi \operatorname{Re} u_*$; $r \operatorname{Im}[\frac{1}{2}g''(a)w_1^2] = 2\pi e_1/(r\bar{D})$, $r \operatorname{Im}[\frac{1}{6}g_3w_1^3] = \alpha_2/r^2$, $\operatorname{Im} e(u_*) = e_1 \operatorname{Im} w + O(\frac{1}{2}E_2\tau^2)$ with $e_1 \operatorname{Im} w_1 = -2\pi e_1/(r\bar{D})$, and $-2\pi \operatorname{Re} u_* = -2\pi a - 2\pi e_1/(r\bar{D}) - 2\pi \operatorname{Re}(w - w_1)$; the three terms $\pm 2\pi e_1/(r\bar{D})$ combine to $-2\pi e_1/(r\bar{D})$, and the remainders are bounded using $|w - w_1| \leq \varepsilon$, $|w|, |w_1| \leq \tau$.) From (8.5): $|F''(u_*)| \geq F_2^- := r\bar{D} - |\operatorname{Re} \kappa_2| - \Theta_2$, $|F''(u_*)| \leq F_2^+ := r\bar{D} + |\kappa_2| + \Theta_2$, and $\arg F''(u_*) = \pi + \psi$ with $|\tan \psi| \leq T := (|\operatorname{Im} \kappa_2| + \Theta_2)/F_2^-$, so $\phi = \pi - \psi/2$, $|\sin \phi| \leq \sin(\frac{1}{2} \arctan T)$, $|\cos \phi| \geq \cos(\frac{1}{2} \arctan T)$.

All these bounds are explicit functions of t ; each of $\tau_1, \eta, \varepsilon, \tau, \Theta_2, \Delta^\pm, \Delta_\alpha, T$ is a sum of positive multiples of powers t^j ($j \geq 1$) and of the increasing functions $M_k(\tau(t))$, $E_k(\tau(t))$ of $\tau(t)$, hence increasing in t , and it suffices to evaluate at $t = t_1$:

$$\begin{array}{lll} \tau_1 = 0.00171431, & N_1 = 35.8379, & N_2 = 3663.095, \\ \bar{\eta} = 4.8915 \cdot 10^{-7}, & L = 1189.92, & L\bar{\eta}/N_2 = 1.589 \cdot 10^{-7}, \\ \varepsilon = 9.7830 \cdot 10^{-7}, & \Theta_2 = 0.02715, & F_2^- = 3665.126, \\ F_2^+ = 3667.090, & T = 5.2635 \cdot 10^{-4}, & |\sin \phi| \leq 2.6317 \cdot 10^{-4}, \\ |\operatorname{Im} u_*| \in [0.00171332, 0.00171528], & \operatorname{Re} u_* \in [4.19294485, 4.19294680], & \Delta^+ = 0.012153, \\ \Delta^- = 0.022930, & \Delta_\alpha = 1.236 \cdot 10^{-5}. & \end{array}$$

(S1): $\operatorname{Re} \text{top} = \operatorname{Re} u_* + s_0 \cos \phi$ lies in $[\operatorname{Re} u_* - s_0, \operatorname{Re} u_* - s_0 \cos(\frac{1}{2} \arctan T)]$, that is in

$$[2.692945, 2.692947] \subset [2.092953, 2.702953],$$

$$\operatorname{Im} \text{top} \in [-|\operatorname{Im} u_*| - s_0|\sin \phi|, -|\operatorname{Im} u_*| + s_0|\sin \phi|] \subset [-0.002110, -0.001319];$$

similarly $\operatorname{Re} \text{bot} \in [5.692945, 5.692947] \subset [5.604839, 20]$, and $\operatorname{Im} \text{bot}$ lies in the same interval as $\operatorname{Im} \text{top}$. So $\text{top} \in B_L$, $\text{bot} \in B_R$. (S2): $\xi_\sigma \geq 2.692945$, $M_3^\sigma s_0/(\frac{3}{2}F_2^-) \leq 0.76721$. (S3): $y_\sigma \geq 1.3186 \cdot 10^{-3} > 0$; multiplying by r , $ry_\sigma \geq \frac{2\pi}{D} - r\varepsilon - s_0 r|\sin \phi|$, and $r\varepsilon, r|\sin \phi|$ are decreasing in r , so (S3) holds for all $r \geq 20001$. (S4): by (8.6) and $\delta \leq F_2^+ s_0^2/8$, it suffices that $r\Gamma \geq B := \Delta^- + (F_2^+ - r\bar{D})s_0^2/8 + S_e - e(a)$ for each row; the right side is at most 0.7287 for all five rows and all $r \geq 20001$ (it is decreasing in r), while $r\Gamma \geq 20001 \cdot 0.06618 > 1324$. (S5): $\operatorname{dist}(\alpha, \pi\mathbb{Z}) \geq \pi \operatorname{dist}(2a, \mathbb{Z}) - \frac{2\pi|e_1|}{r\bar{D}} - \frac{|\alpha_2|}{r^2} - \Delta_\alpha \geq 1.21236 - 0.000043 - 0.000001 - 1.2 \cdot 10^{-5} > 1.21$. (S6): $-H_d \geq -r\hat{g}(a) - e(a) - \Delta^+$, so $-H_d - (\bar{c}_2 r - 3) \geq (17.385899)r + 12.3453 > 0$.

8.8. Proof of Theorem 8.1. For $r \leq 15$ see the beginning of this section. For $17 \leq r \leq 49$, Table 6 certifies $\boldsymbol{\eta}^{(r)}$ by the programme of Section 5 with the path of §5.1 (and Y doubled until the tails pass, as in Section 7), except that (P4) is verified through Lemma 8.3: the dominant mode on the lower-half-plane pieces and on the upper ray, the mode $k = 1$ on the upper ray, and $y_\sigma > 0$; the certificates are produced by `certify2.py`, which evaluates the phase function with the seven distinct directions and their multiplicities. For $51 \leq r \leq 19999$, §8.6 verifies (S1)–(S6); for $r \geq 20001$, §8.7 does. Proposition 8.8 concludes. $\square$

Remark 8.9. (i) The same argument applies to any family $\boldsymbol{\eta}^{(r)} = (\eta_0; z^r; p^{c_p r} (p \in \mathcal{P}); \dots)$ with $c_p \in \mathbb{N}$, finitely many fixed extra directions, and $C = \sum_p c_p$; it gives “one of $\zeta(r+2), \dots, \zeta((C+1)r+c)$ ” for all large r as soon as $-\hat{g}(a) > \hat{m}_1 + \sum_p c_p \hat{m}(p) - \bar{I}_\Phi$; the smallest admissible integer is $C = 5$ (for $C = 4$ the criterion fails, §7.1), which is why the constant is 6. With rational weights $c_p \in \frac{1}{4}\mathbb{Z}$ and multiplicities $c_p k_r$, where k_r is the even integer among $\frac{r+1}{4}, \frac{r+3}{4}$, the perturbation e becomes

TABLE 6. Certificates for $\eta^{(r)}$, $17 \leq r \leq 49$ ($q = 6r + 1$). $\theta_{\min}$ is the smallest value of $\theta - \sup H$ over the pieces $[\text{top}, x_0]$, $[\text{ext}, \text{bot}]$, both rays (dominant mode) and the upper ray for $k = 1$, with $\theta = H_d - \delta/2$; the tail bounds are below θ by at least 199 in all cases.

r	q	u_*	C_0	C_2	$C_0 - C_2$	s_0	δ	y_σ	$\text{dist}(\alpha, \pi\mathbb{Z})$	$\theta_{\min}$
17	103	$3.617606 - 1.853535i$	3960.1956	3616.9557	+343.24	1.179	0.580	1.510	0.064	0.70
19	115	$3.724571 - 1.685226i$	4423.6932	4046.2749	+377.42	1.190	0.652	1.372	0.176	2.00
21	127	$3.804581 - 1.542932i$	4887.2933	4475.5941	+411.70	1.198	0.724	1.256	0.355	2.22
23	139	$3.865883 - 1.421536i$	5350.9724	4904.9132	+446.06	1.205	0.795	1.157	0.490	0.74
25	151	$3.913828 - 1.317027i$	5814.7130	5334.2324	+480.48	1.210	0.867	1.071	0.596	2.65
27	163	$3.952001 - 1.226275i$	6278.5027	5763.5516	+514.95	1.213	0.938	0.997	0.680	2.86
29	175	$3.982870 - 1.146838i$	6742.3320	6192.8708	+549.46	1.217	1.009	0.933	0.748	3.08
31	187	$4.008173 - 1.076792i$	7206.1936	6622.1899	+584.00	1.219	1.080	0.876	0.803	3.30
33	199	$4.029167 - 1.014610i$	7670.0819	7051.5091	+618.57	1.221	1.150	0.825	0.849	3.51
35	211	$4.046773 - 0.959070i$	8133.9927	7480.8283	+653.16	1.223	1.221	0.780	0.887	3.72
37	223	$4.061680 - 0.909183i$	8597.9224	7910.1475	+687.77	1.225	1.292	0.739	0.920	3.94
39	235	$4.074411 - 0.864144i$	9061.8681	8339.4666	+722.40	1.226	1.363	0.703	0.948	4.15
41	247	$4.085369 - 0.823290i$	9525.8277	8768.7858	+757.04	1.227	1.433	0.669	0.972	4.37
43	259	$4.094868 - 0.786073i$	9989.7993	9198.1050	+791.69	1.228	1.504	0.639	0.992	4.58
45	271	$4.103155 - 0.752034i$	10453.7812	9627.4241	+826.36	1.229	1.574	0.611	1.011	4.79
47	283	$4.110429 - 0.720788i$	10917.7723	10056.7433	+861.03	1.230	1.645	0.586	1.027	3.47
49	295	$4.116846 - 0.692007i$	11381.7714	10486.0625	+895.71	1.231	1.716	0.562	1.041	5.22

one of four fixed functions $\frac{\theta}{4} \sum_p c_p \Pi_p$ (with $\theta \in \{-1, 1, 3, 5\}$), and the same proof gives the constant $1 + C/4$ for any $C/4 > \lambda_* \approx 4.15$, at the price of larger constants and a larger r_0 . We have not carried this out. (ii) The proof shows more precisely that for $\eta^{(r)}$, $C_0 - C_2(r) \geq 17.38r + 12$: the linear forms are far from optimal, and the constant 6 is a convenient integer, not the limit of the method.

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